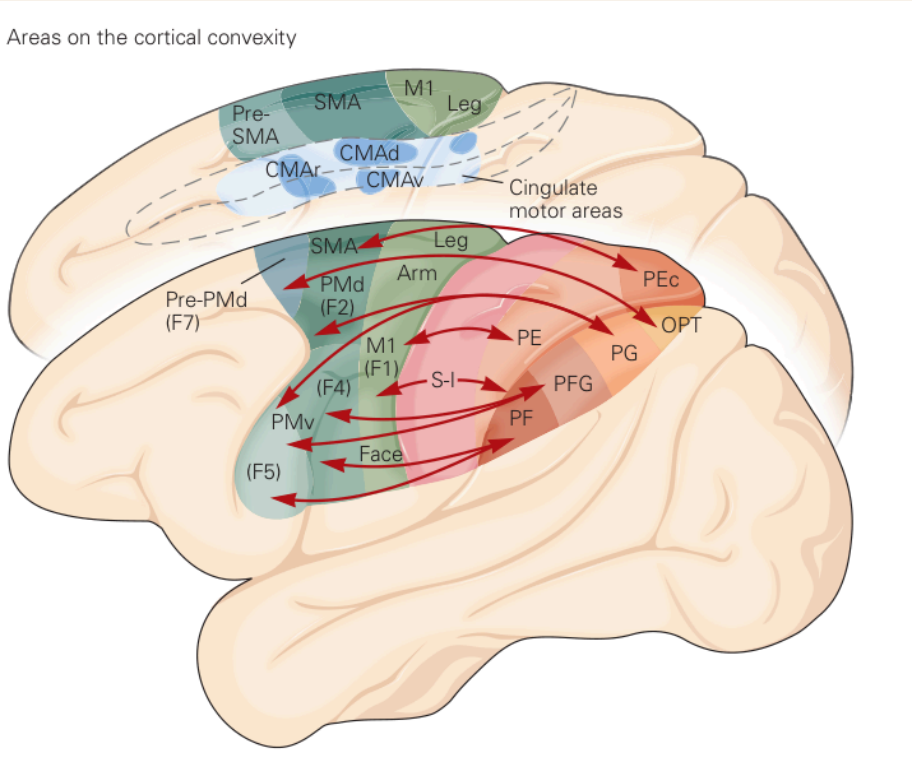
PHYSIOLOGY OF THE MOTOR SYSTEM

From an integrated body–environment view to action

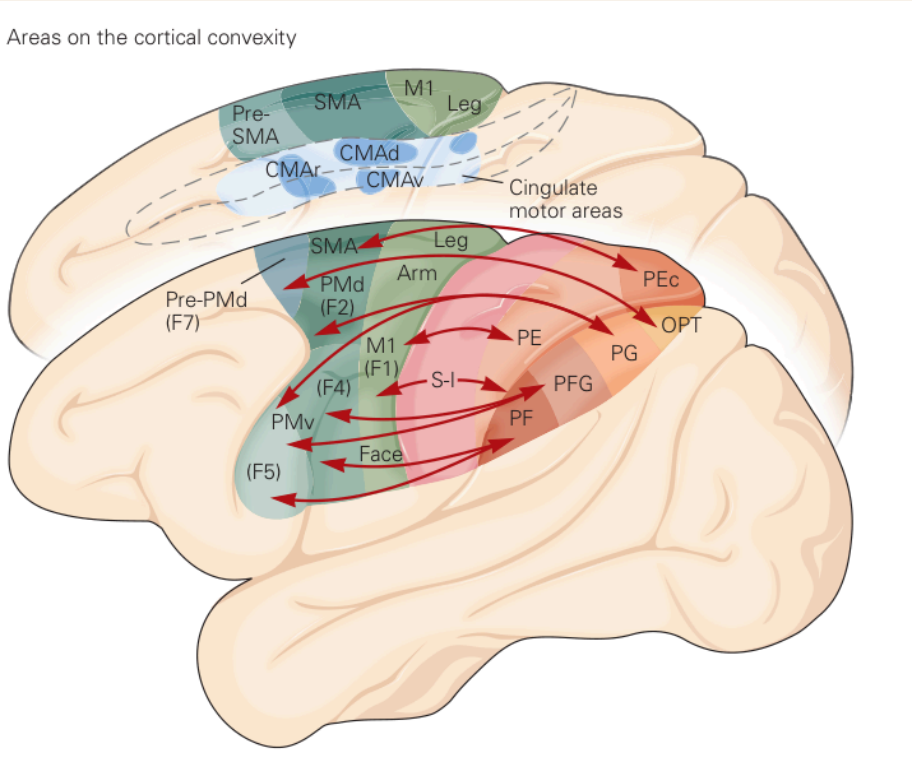
Lecture 02

A connected cortical network



Kandel 6e, Fig. 34–2B, p. 820 · macaque cortical convexity

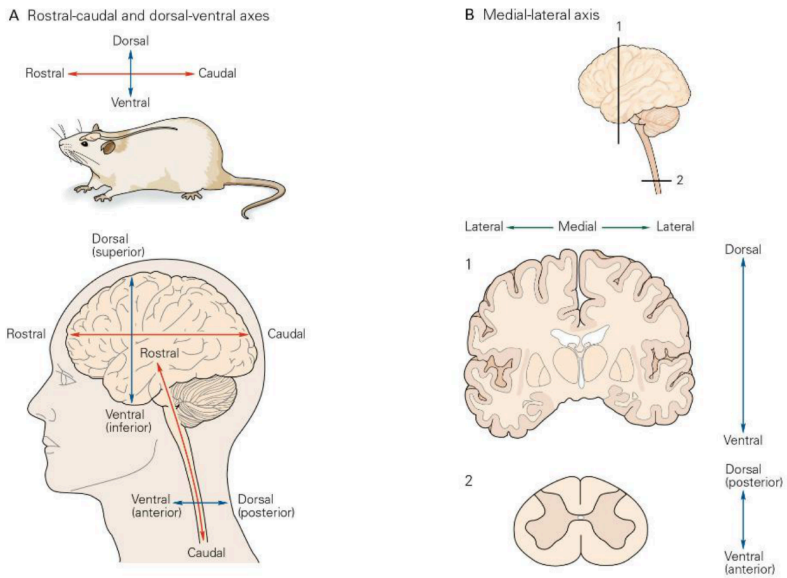
A connected cortical network



Kandel 6e, Fig. 34–2B, p. 820 · macaque cortical convexity

> The cortical motor network includes parietal, premotor and primary motor regions.

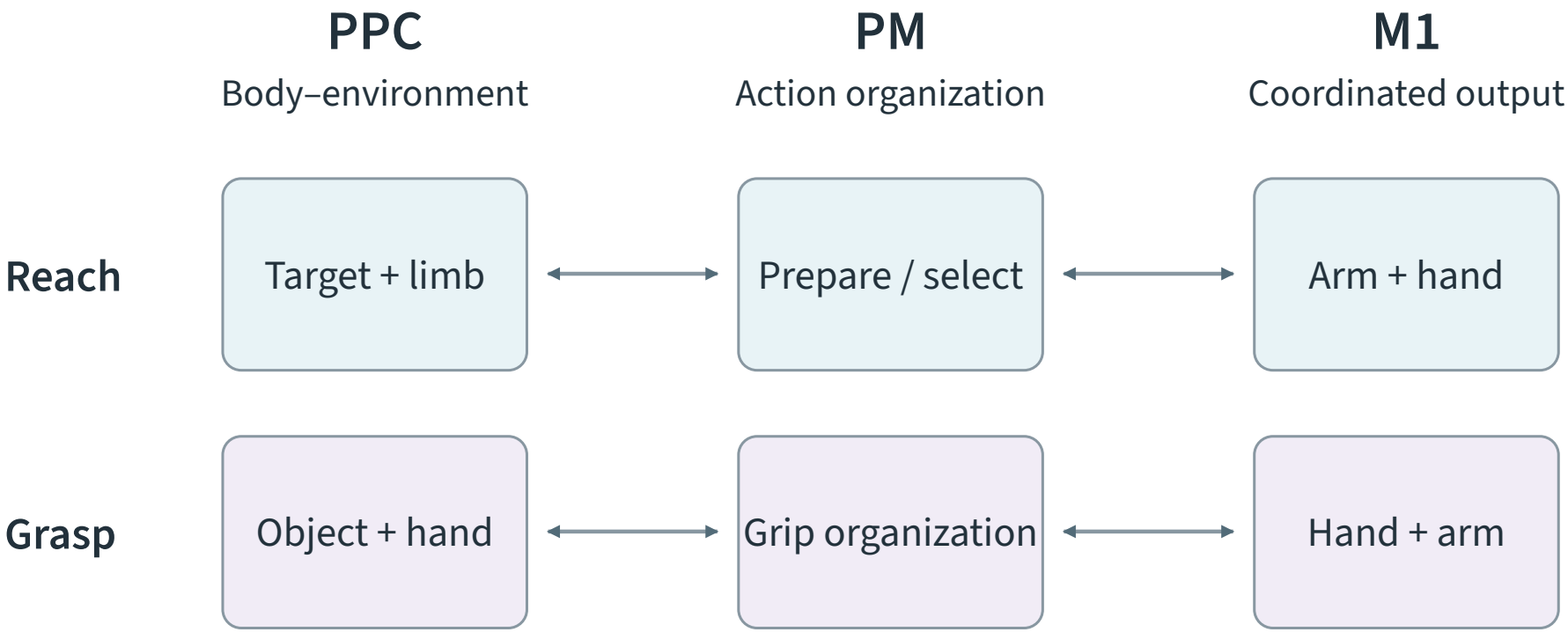
Read the anatomical directions



Prior L01 direction slide · Kandel Fig. 1–2 · archived 2025–2026 deck, p. 26

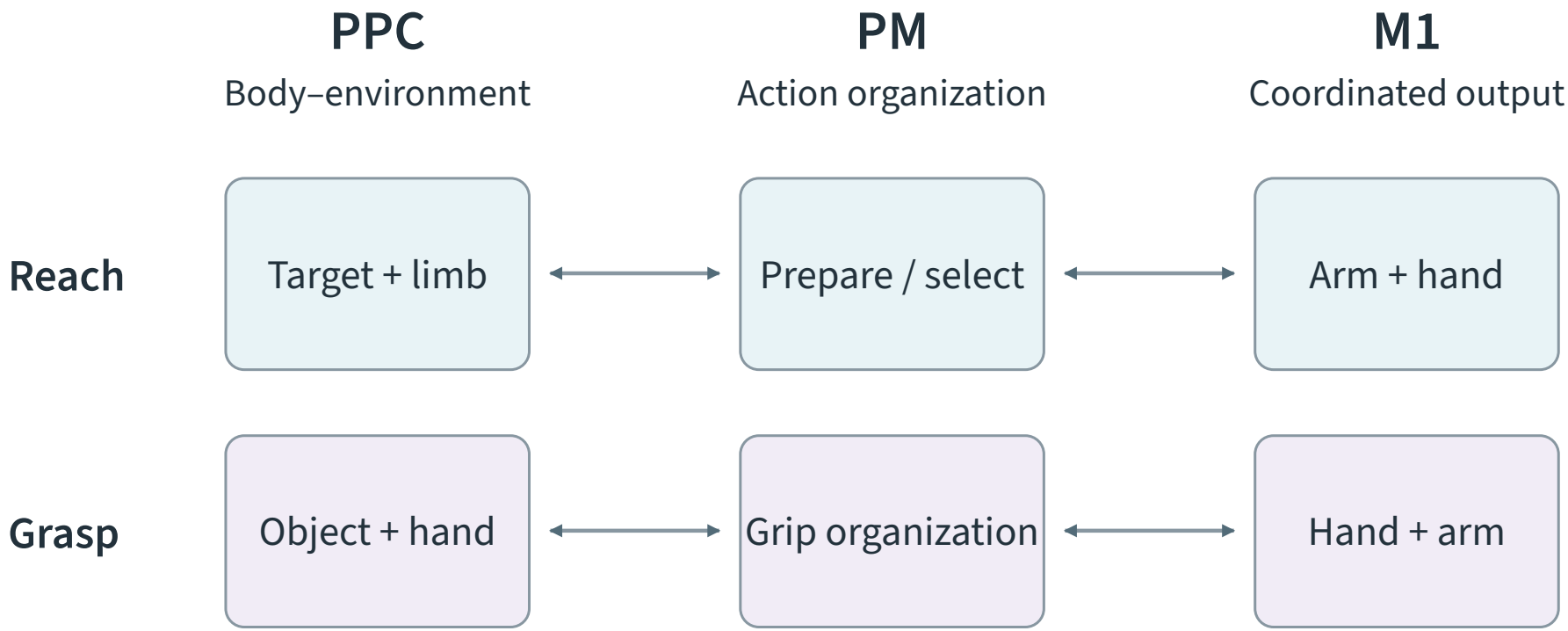
- Rostral: towards the nose / Caudal: towards the tail
- Medial: towards the midline / Lateral: away from the midline
- Dorsal: towards the back / Ventral: towards the belly

From body–environment relationships to action



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

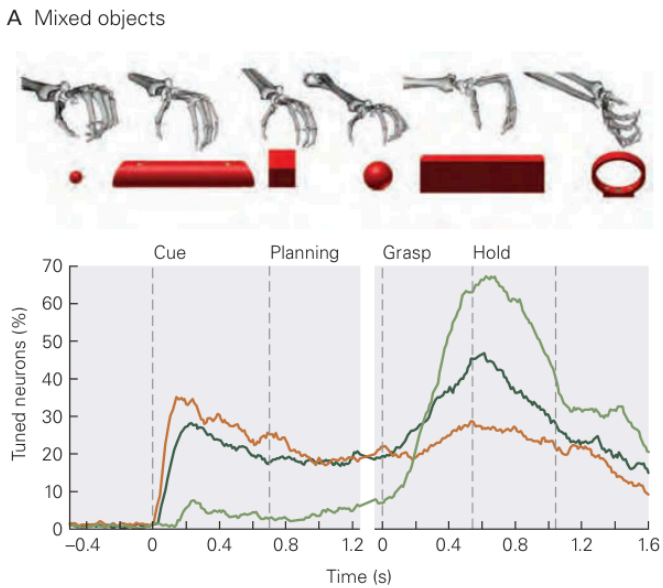
From body–environment relationships to action



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

- Physiological distinctions between processing stage and movement type.
- Interacting contributions, not a compulsory relay.

Which information differs across the grasping circuit?

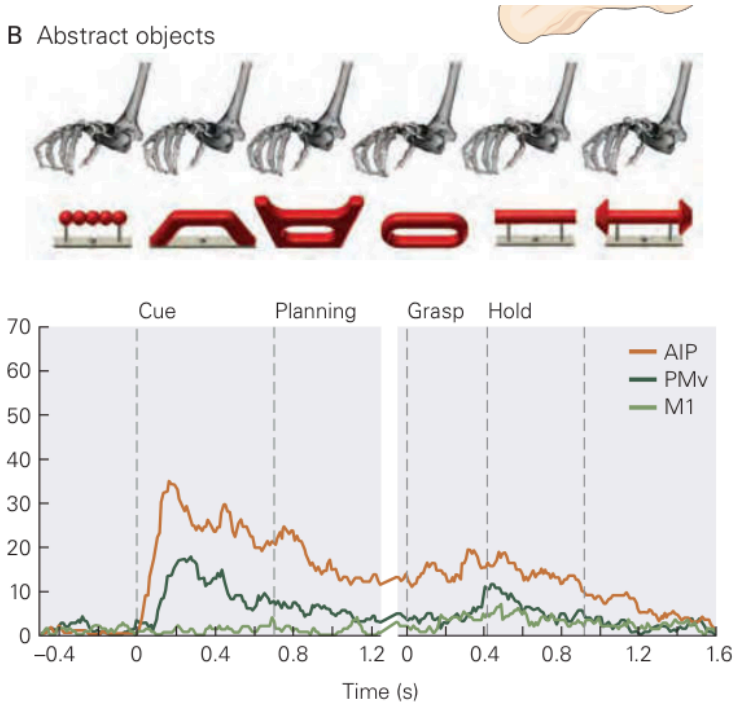


AIP
PMv
M1

Kandel 6e, Fig. 34–15A, p. 840 · mixed objects; original shared legend

- Cue: Object appears
- Planning: Prepare the grasp
- Grasp: Reach, then grasp the object
- Hold: Maintain the grasp

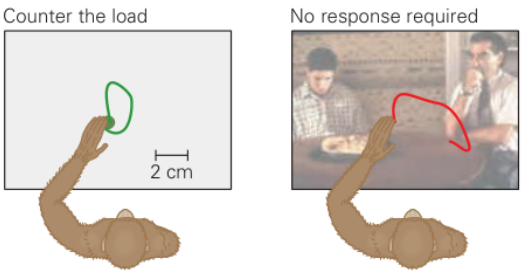
Different objects can require similar grasps



Kandel 6e, Fig. 34–15B, p. 840 · abstract objects

- Different shapes, similar grips

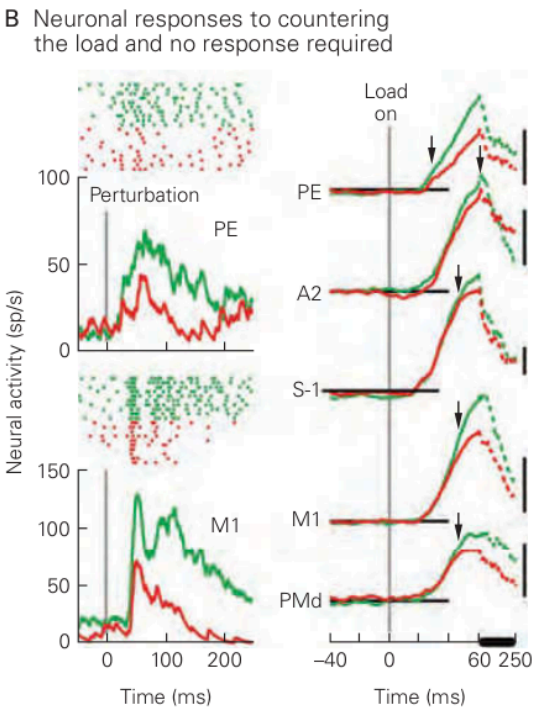
Task engagement changes feedback responses

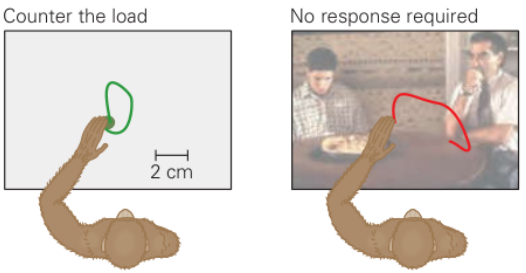


- Unexpected arm load
- Green: correction to center
- Red: movie; no correction required

Kandel 6e, Fig. 34–23A/B, p. 853 · counter the load versus no response

Task engagement changes feedback responses





- Unexpected arm load
- Green: correction to center
- Red: movie; no correction required

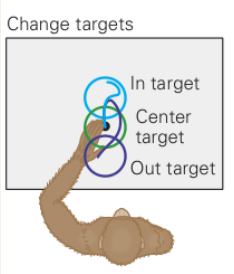
- Feedback responses across regions
- Earliest task modulation: PE
- Arrows: first condition difference

Kandel 6e, Fig. 34–23A/B, p. 853 · counter the load versus no response

PE: superior parietal area 5
A2: somatosensory area 2
S1: primary somatosensory areas 1 and 3

M1: primary motor cortex
PMd: dorsal premotor cortex

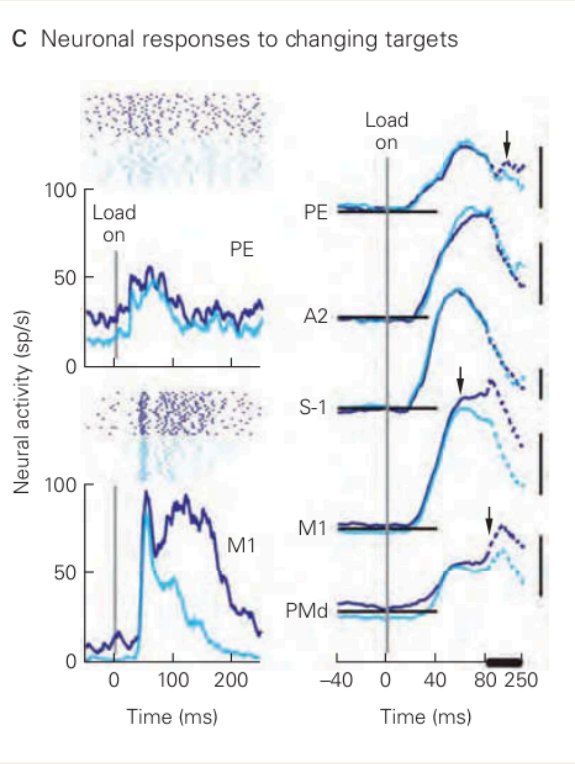
Corrective demands change the regional timing

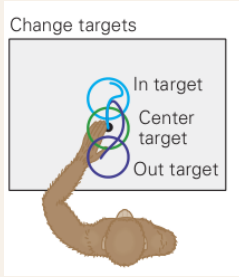


- Arm load: cue to a new target
- Cyan: push toward target (IN)
- Purple: push away (OUT)

Kandel 6e, Fig. 34–23A/C, p. 853 · disturbance toward versus away from the new target

Corrective demands change the regional timing





- Arm load: cue to a new target
- Cyan: push toward target (IN)
- Purple: push away (OUT)

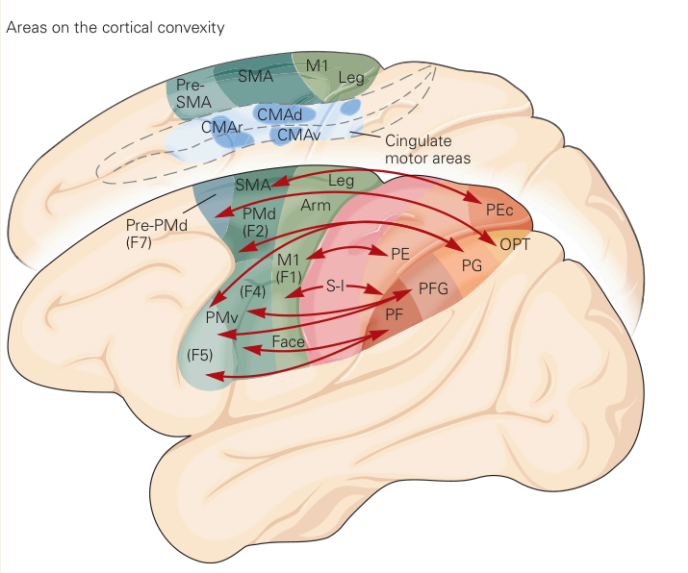
- Earlier target modulation in M1
- Later parietal target differences
- Timing dependent on the comparison

Kandel 6e, Fig. 34–23A/C, p. 853 · disturbance toward versus away from the new target

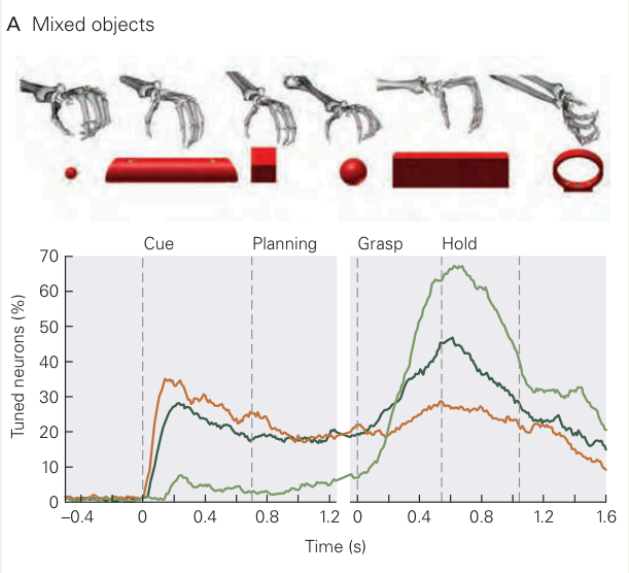
PE: superior parietal area 5
A2: somatosensory area 2
S1: primary somatosensory areas 1 and 3

M1: primary motor cortex
PMd: dorsal premotor cortex

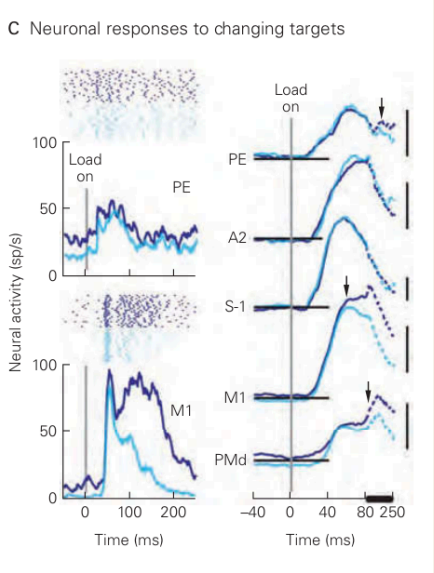
Section A summary: one network, shifting emphases



Connected network



Shifting emphasis

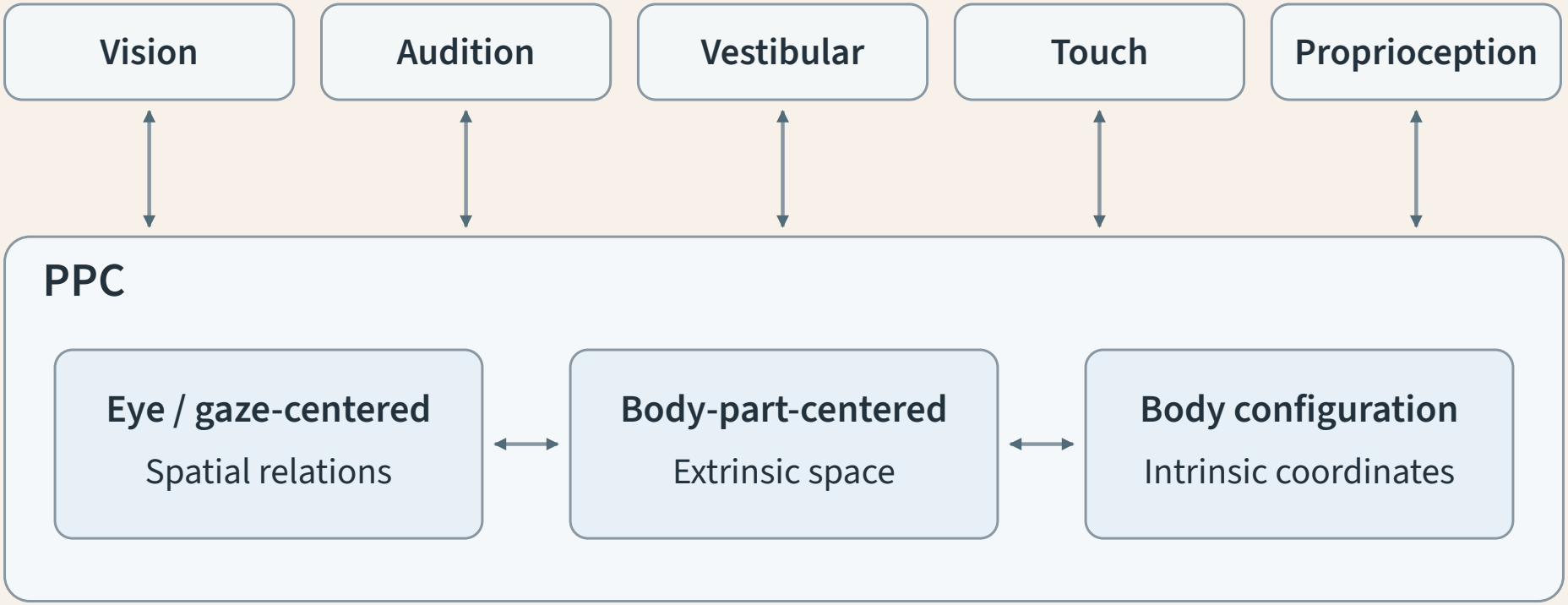


Task-dependent feedback

TAKE HOME One connected network; regional emphasis depends on the task.

Kandel 6e, Figs. 34–2B, p. 820; 34–15A, p. 840; 34–23C, p. 853

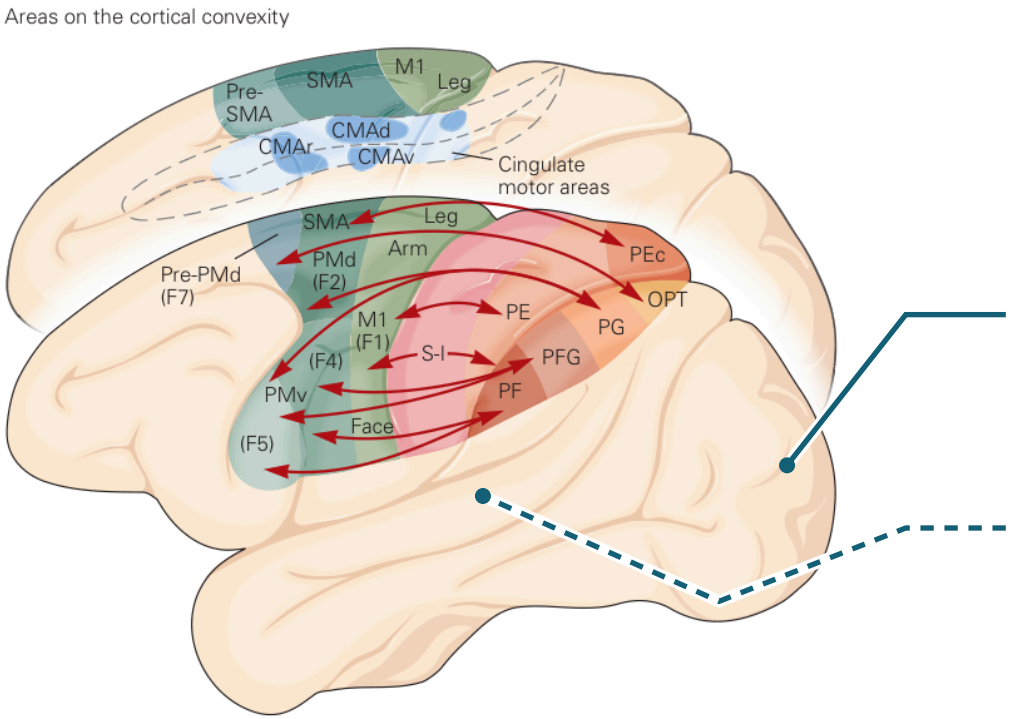
B · Integrating body and environment



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

- Parietal organization reflects the organization of its sensory inputs.
- Linked, mixed representations.

Locate the sensory sources around PPC



Visual cortex

Posterior occipital region

Auditory cortex

Superior temporal plane

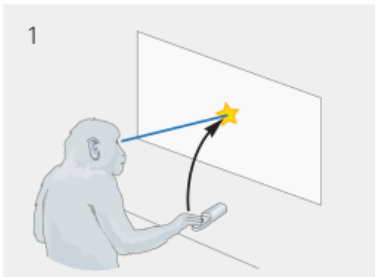
Largely buried in the sulcus

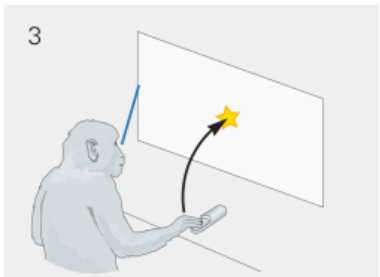
Approximate regional locators

Kandel 6e, Fig. 34–2B, p. 820 · added approximate visual/auditory locators; auditory region largely buried.

> Posterior parietal cortex integrates different sensory signals to identify possible targets of action.

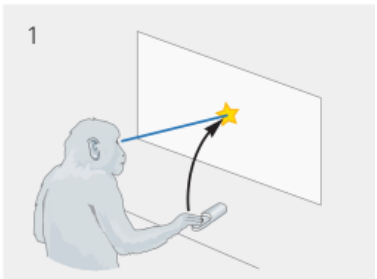
Same reach, different gaze

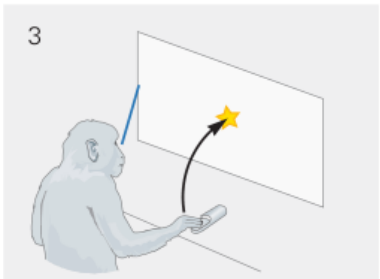




Kandel 6e, Fig. 34–5C, conditions 1 and 3, pp. 826–827 · task drawings only; geometry illustrative

Same reach, different gaze





1 · Gaze at target

H — G = T

Target from gaze: T − G = 0

Hand from gaze: H − G

Reach: T − H

3 · Gaze elsewhere

G — H — T

Target from gaze: T − G

Hand from gaze: H − G

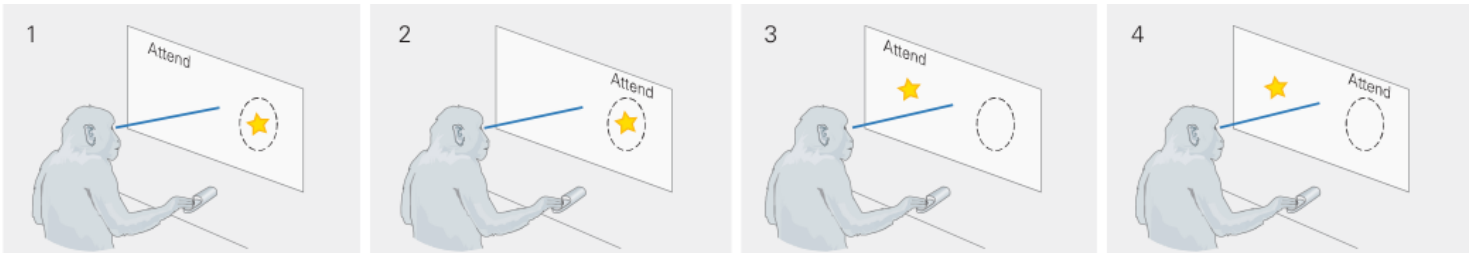
Reach: T − H

$$(T - G) - (H - G) = T - H$$

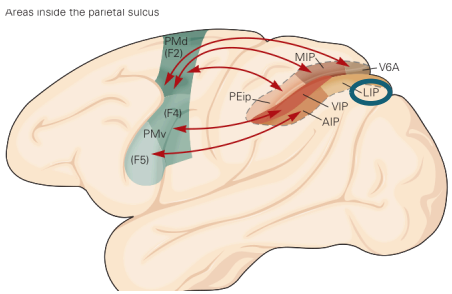
G: gaze · H: hand · T: target

Kandel 6e, Fig. 34–5C, conditions 1 and 3, pp. 826–827 · task drawings only; geometry illustrative

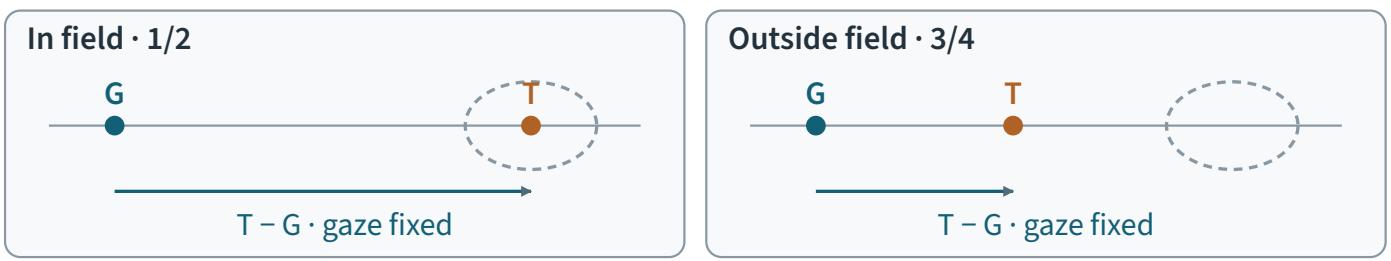
LIP · visual location and attention



Lateral intraparietal area



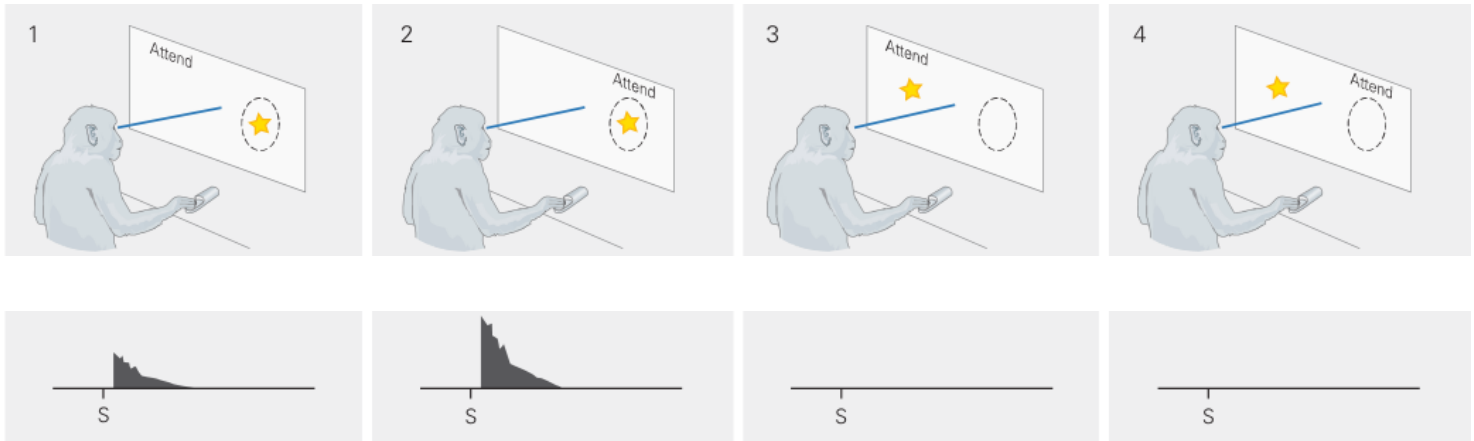
- Dotted circle: field
- 1 ↔ 2: attention
- 1/2 ↔ 3/4: location



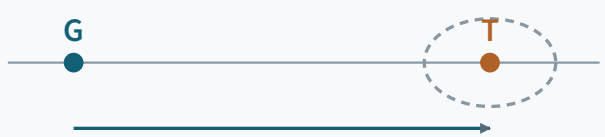
G: gaze · T: flashed stimulus · dotted oval: receptive field

Kandel 6e, Figs. 34–5A, pp. 826–827; 34–2B, p. 820 · schematic illustrative

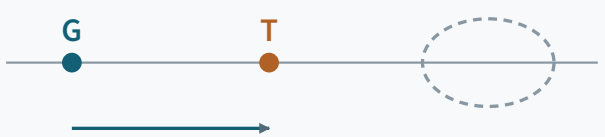
LIP · visual location and attention



In field · 1/2



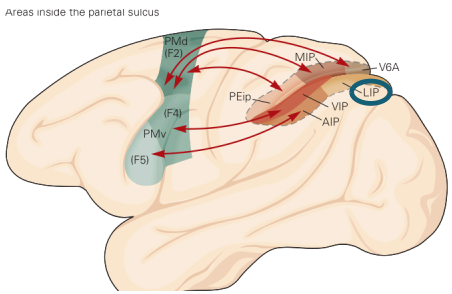
Outside field · 3/4



G: gaze · T: flashed stimulus · dotted oval: receptive field

Kandel 6e, Figs. 34–5A, pp. 826–827; 34–2B, p. 820 · schematic illustrative

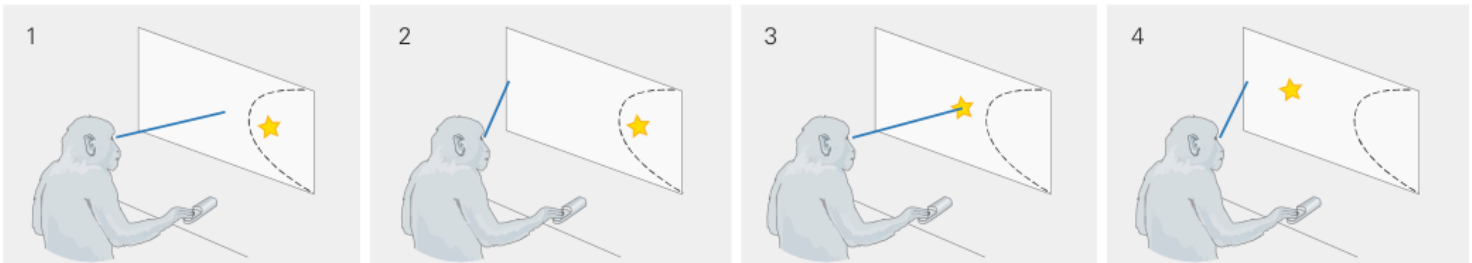
Lateral intraparietal area



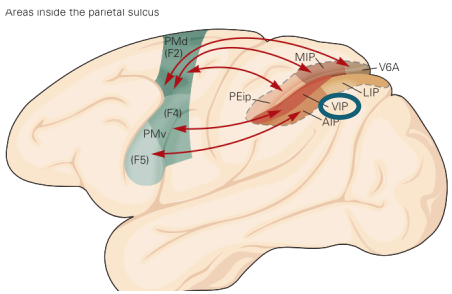
- Dotted circle: field
- 1 ↔ 2: attention
- 1/2 ↔ 3/4: location

- In field: response
- Attended: stronger
- Outside: little response

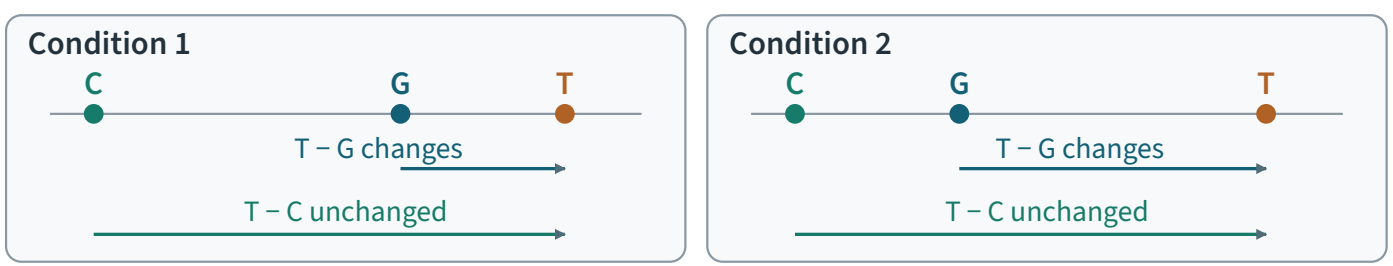
VIP · location relative to the head



Ventral intraparietal area



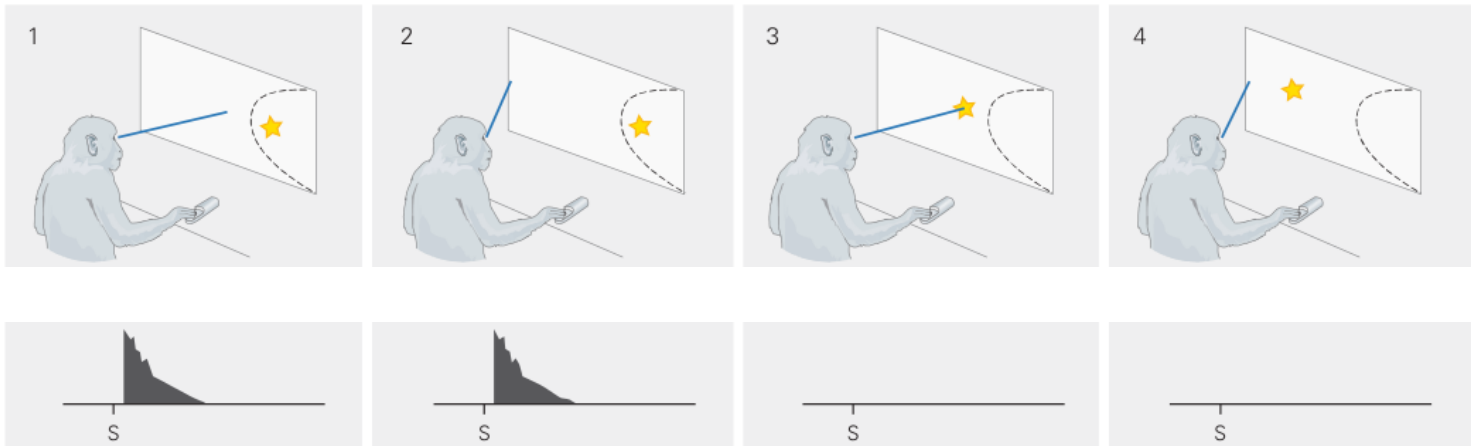
- 1 ↔ 2: same head-relative location
- 1 ↔ 4: similar retinal offset

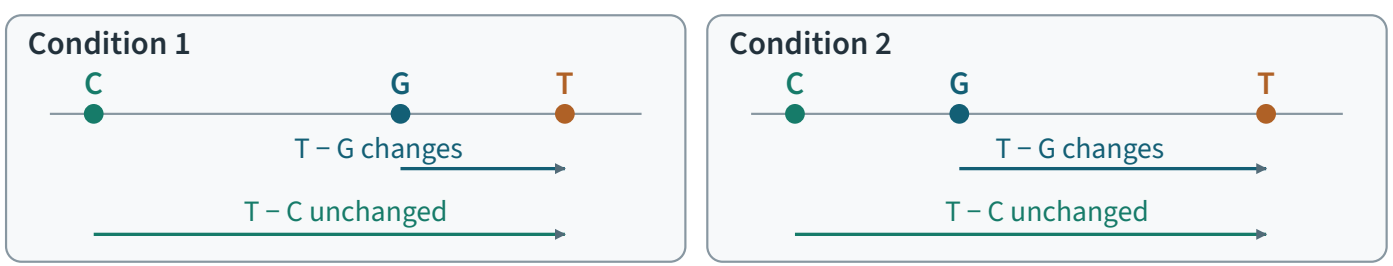


C: head · G: gaze · T: flashed stimulus · geometry of 1/2

Kandel 6e, Figs. 34–5B, pp. 826–827; 34–2B, p. 820 · schematic illustrative

VIP · location relative to the head

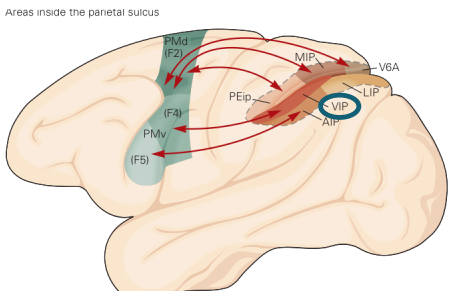




C: head · G: gaze · T: flashed stimulus · geometry of 1/2

Kandel 6e, Figs. 34–5B, pp. 826–827; 34–2B, p. 820 · schematic illustrative

Ventral intraparietal area

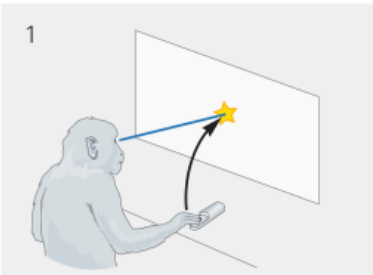


- 1 ↔ 2: same head-relative location
- 1 ↔ 4: similar retinal offset

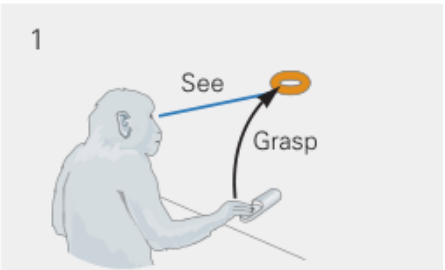
- 1/2: similar response
- 1/4: different response
- Head-relative sensitivity

Reach and grasp: the hand acts on the target

REACH



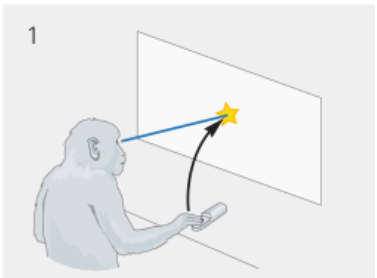
GRASP



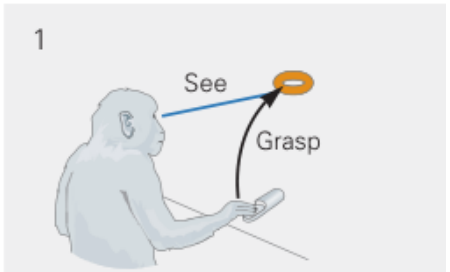
Kandel 6e, Fig. 34–5C and D, condition 1, pp. 826–827 · task drawings only; geometry illustrative

Reach and grasp: the hand acts on the target

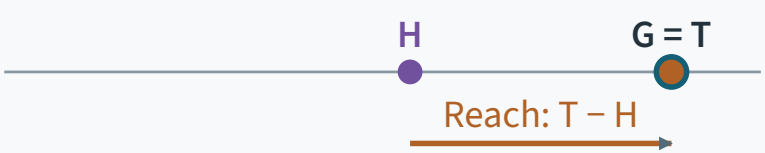
REACH



GRASP

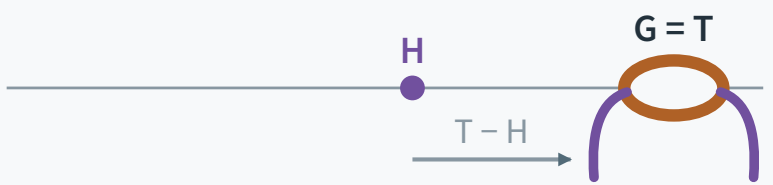


Reach — MIP example



Where the hand goes

Grasp — AIP example



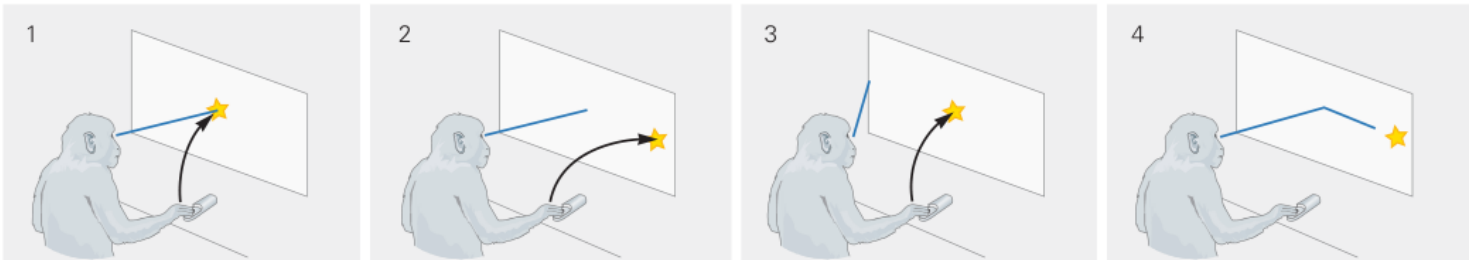
How the hand is shaped

G: gaze · H: hand · T: target or object · illustrative geometry

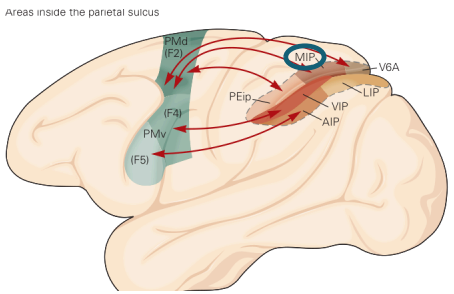
> Acting needs the target relative to the hand, and the hand shaped to the object.

Kandel 6e, Fig. 34–5C and D, condition 1, pp. 826–827 · task drawings only; geometry illustrative

MIP · reach direction relative to gaze



Medial intraparietal area



- 1 ↔ 3: same reach, different gaze
- 2 ↔ 3: different reach
- 4: eye movement, no reach

1 · Gaze at target

H — G = T

T − G = 0

T − H: same reach

3 · Gaze elsewhere

G — H — T

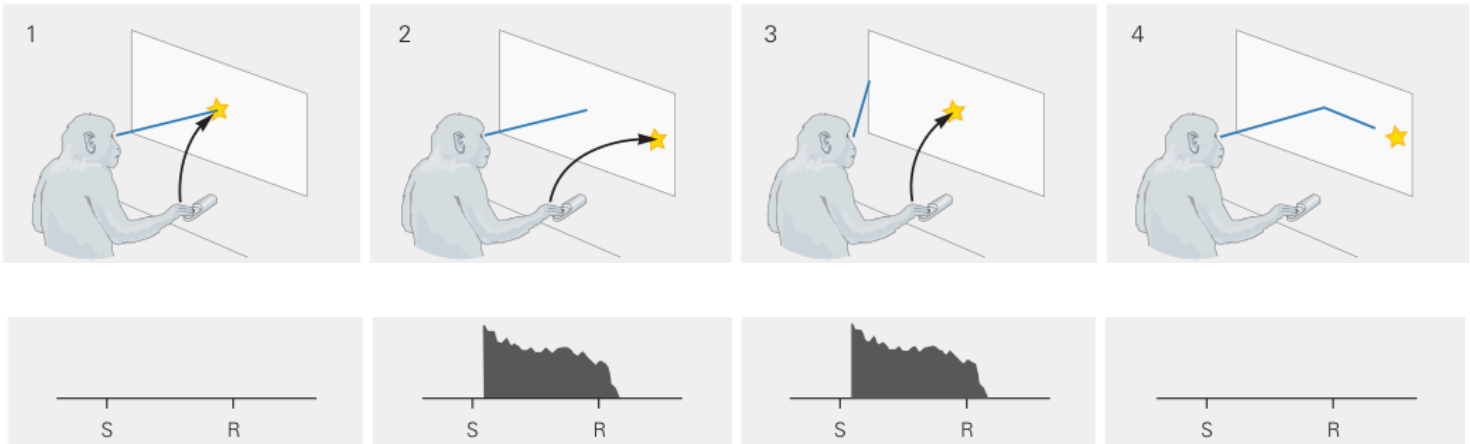
T − G: target right of gaze

T − H: same reach

G: gaze · H: hand · T: reach target · geometry of 1/3

Kandel 6e, Figs. 34–5C, pp. 826–827; 34–2B, p. 820 · schematic illustrative

MIP · reach direction relative to gaze



1 · Gaze at target

H G = T

T − G = 0

T − H: same reach

3 · Gaze elsewhere

G H T

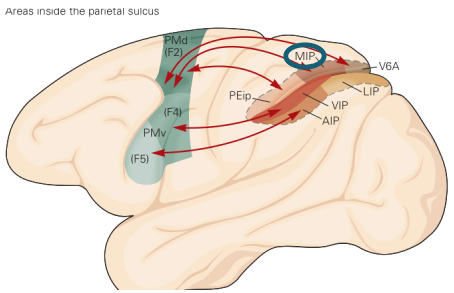
T − G: target right of gaze

T − H: same reach

G: gaze · H: hand · T: reach target · geometry of 1/3

Kandel 6e, Figs. 34–5C, pp. 826–827; 34–2B, p. 820 · schematic illustrative

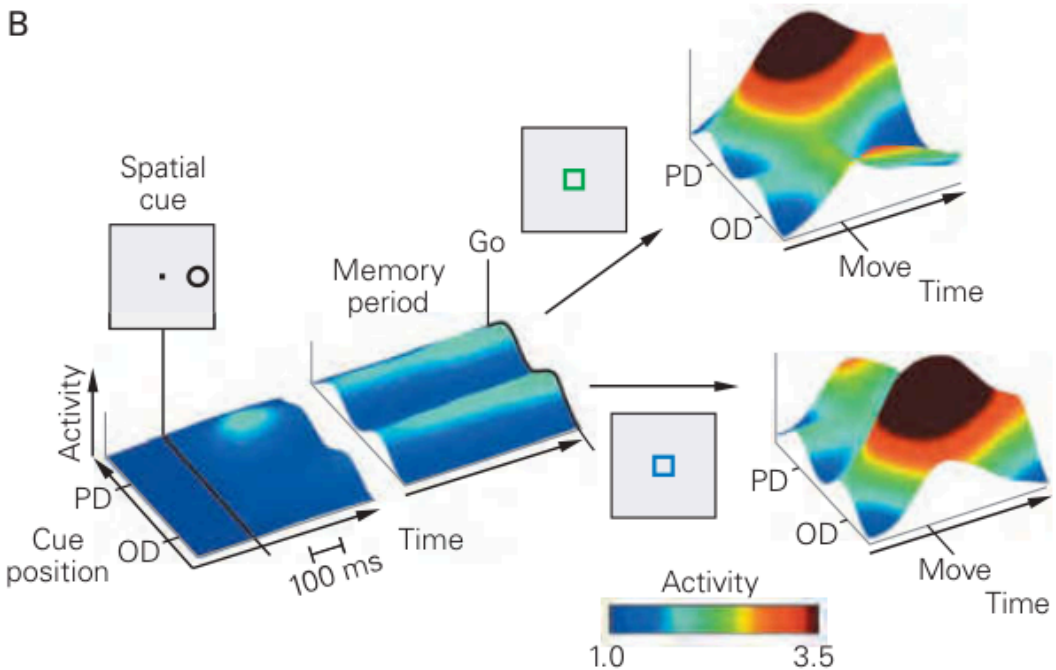
Medial intraparietal area



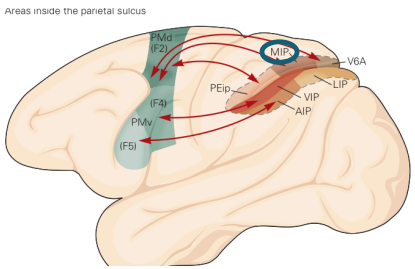
- 1 ↔ 3: same reach, different gaze
- 2 ↔ 3: different reach
- 4: eye movement, no reach

- Only 2 and 3 respond
- Both: target right of gaze
- No reach: no response

PRR · potential reaches in parietal cortex



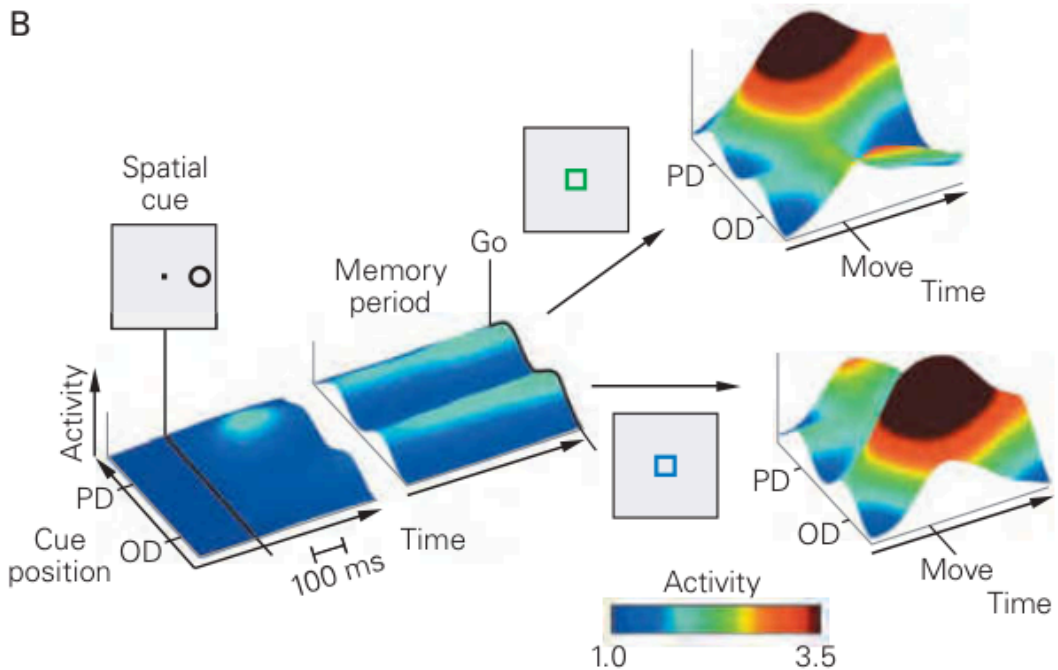
Parietal reach region (≈ MIP)



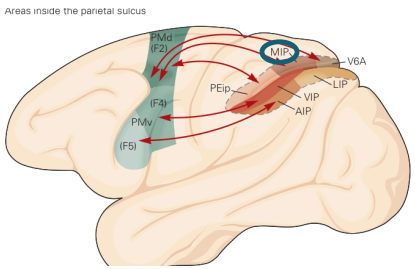
- One spatial cue
- Later color cue: reach to it (PD) or opposite (OD)

Kandel 6e, Fig. 34–8B, p. 832; PRR likely corresponds to MIP, p. 825; 34–2B, p. 820 · remembered target and later reach instruction

PRR · potential reaches in parietal cortex



Parietal reach region (≈ MIP)



- One spatial cue
- Later color cue: reach to it (PD) or opposite (OD)

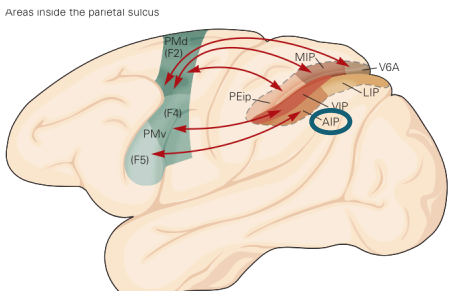
> - Delay: both directions present
> - Then the instructed reach
> - Options held before choice

Kandel 6e, Fig. 34–8B, p. 832; PRR likely corresponds to MIP, p. 825; 34–2B, p. 820 · remembered target and later reach instruction

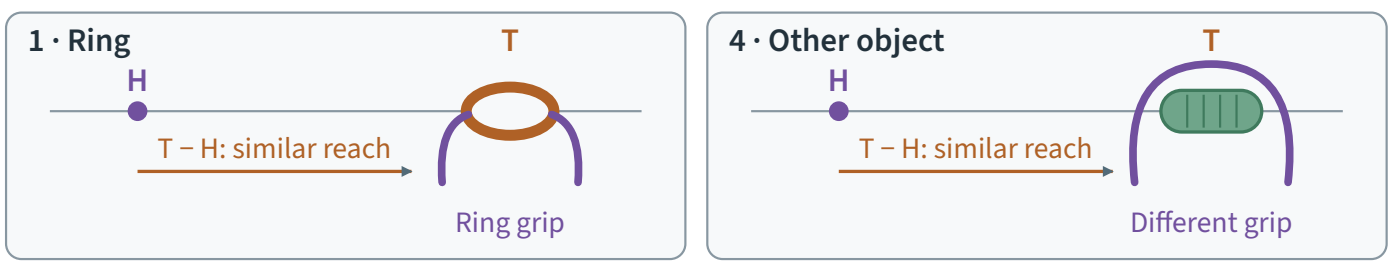
AIP · object shape for grasping



Anterior intraparietal area



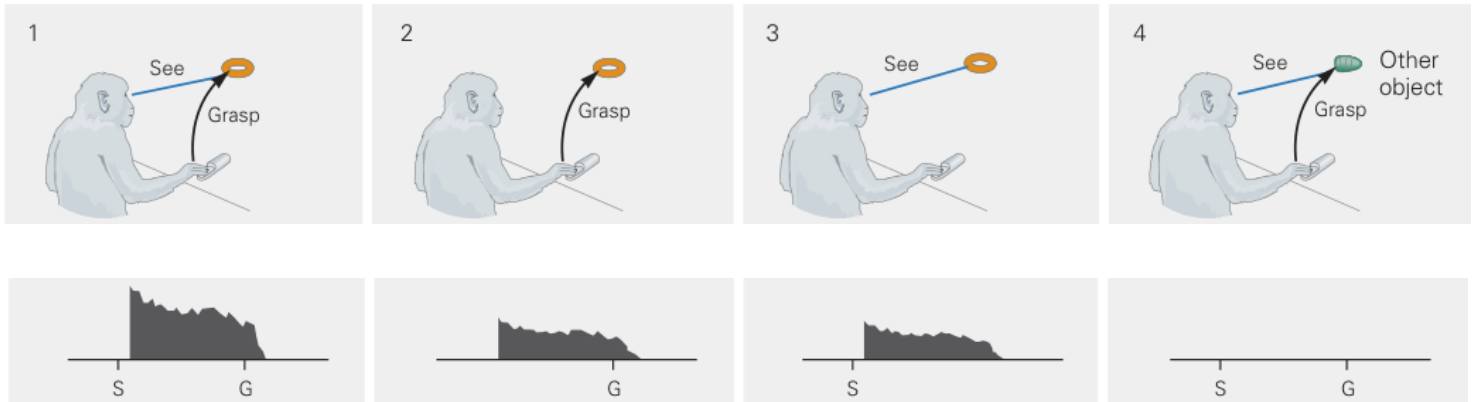
- 1 ↔ 4: similar reach, other object
- 2: grasp in darkness
- 3: view, no grasp

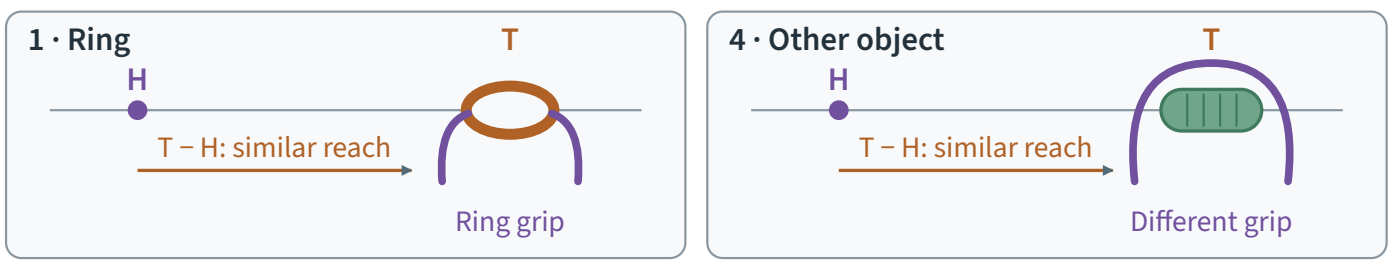


H: hand · T: object location · similar position, different shape and grip

Kandel 6e, Figs. 34–5D, pp. 826–827; 34–2B, p. 820 · schematic illustrative

AIP · object shape for grasping

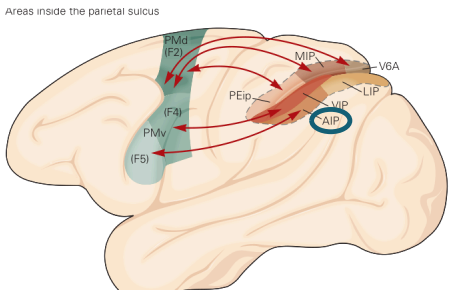




H: hand · T: object location · similar position, different shape and grip

Kandel 6e, Figs. 34–5D, pp. 826–827; 34–2B, p. 820 · schematic illustrative

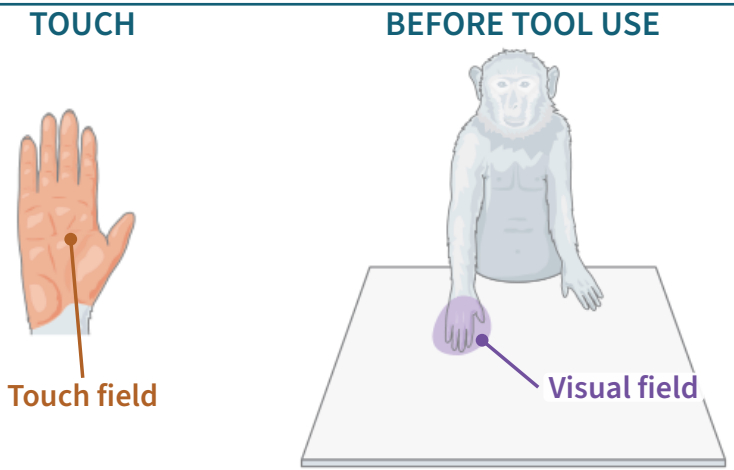
Anterior intraparietal area



- 1 ↔ 4: similar reach, other object
- 2: grasp in darkness
- 3: view, no grasp

- Other object: none
- Dark grasp: response
- View only: response

Parietal cortex · visual space around the hand

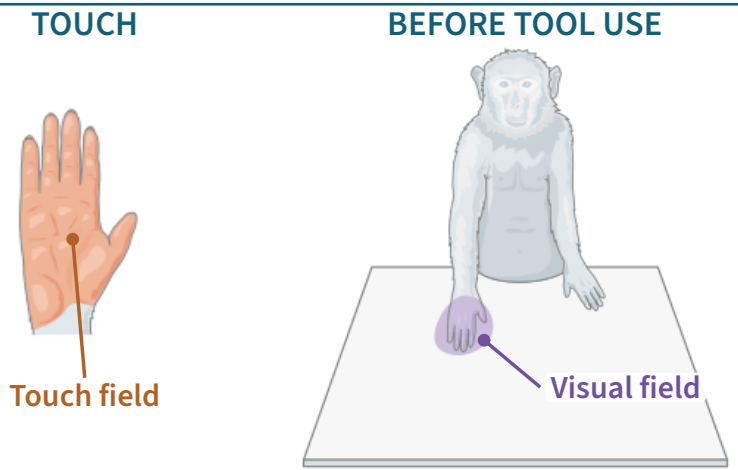


Hand-centered example

- Touch field on the hand
- Visual field near the hand

Kandel 6e, Fig. 34–6A, p. 828 · receptive-field summary; companion geometry illustrative

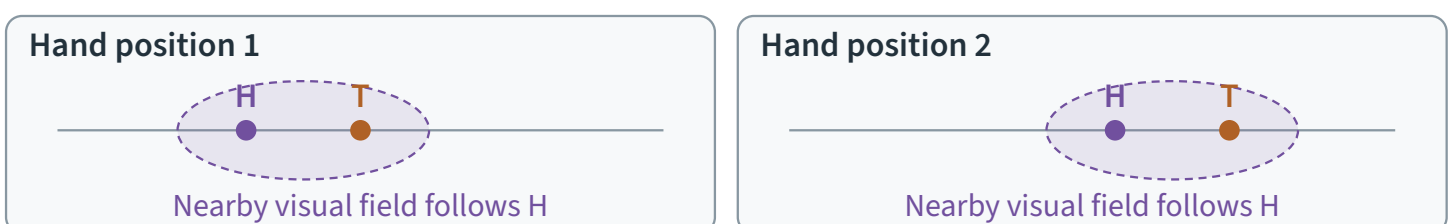
Parietal cortex · visual space around the hand



Hand-centered example

- Touch field on the hand
- Visual field near the hand

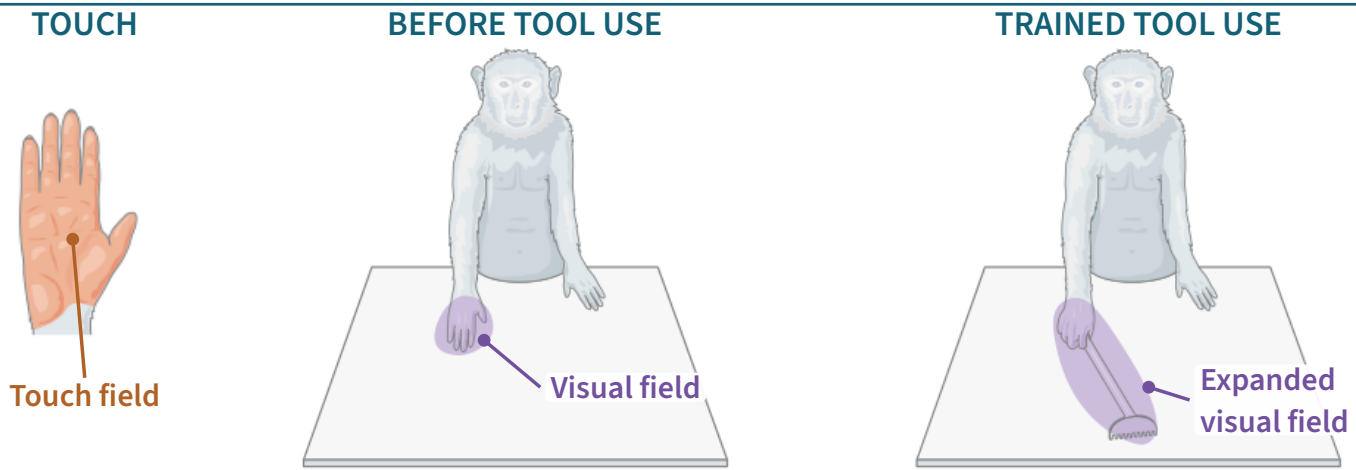
> Anchor: the hand



H: hand · T: visual probe · explanatory field, not measured boundaries

Kandel 6e, Fig. 34–6A, p. 828 · receptive-field summary; companion geometry illustrative

Parietal cortex · visual space around the hand

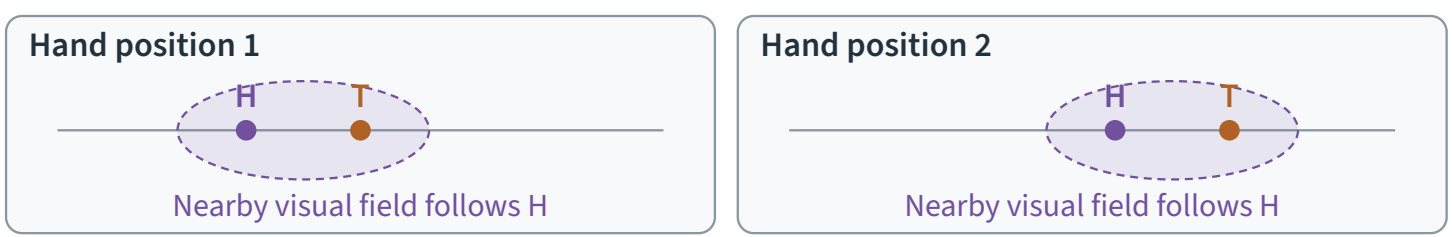


Hand-centered example

- Touch field on the hand
- Visual field near the hand

Anchor: the hand

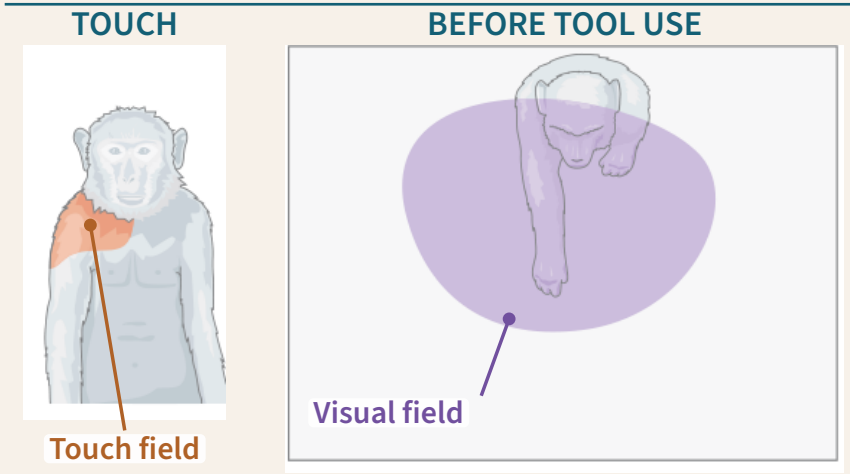
- Learned tool use
- Expanded visual field
- Hand remains the anchor



H: hand · T: visual probe · explanatory field, not measured boundaries

Kandel 6e, Fig. 34–6A, p. 828 · receptive-field summary; companion geometry illustrative

Parietal cortex · a shoulder-centered workspace

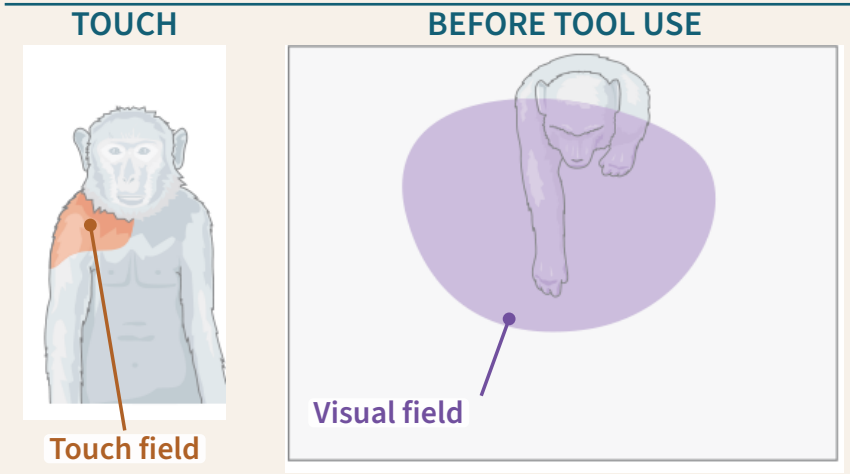


Shoulder-centered example

- Touch field at the shoulder
- Larger nearby visual field

Kandel 6e, Fig. 34–6B, p. 828 · a different bimodal neuron; receptive-field summary; companion geometry illustrative

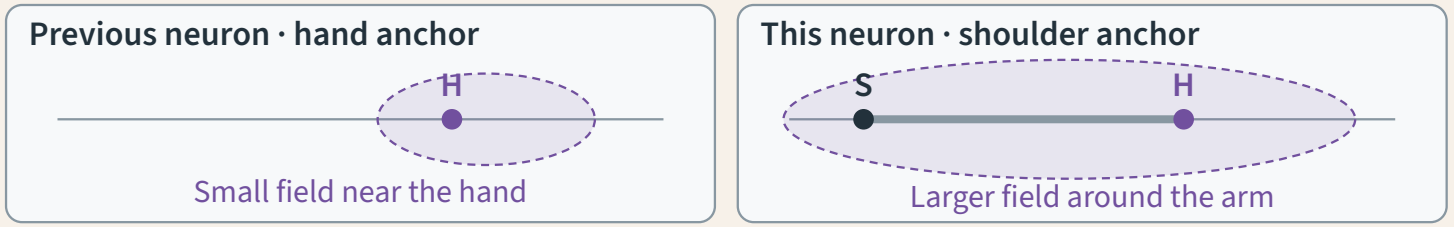
Parietal cortex · a shoulder-centered workspace



Shoulder-centered example

- Touch field at the shoulder
- Larger nearby visual field

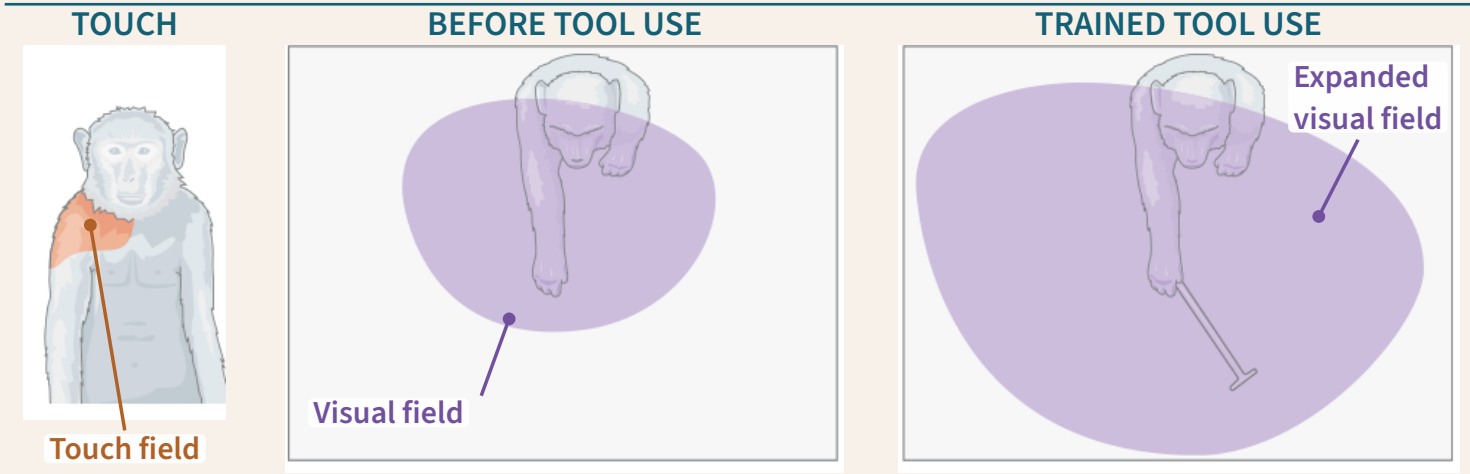
> Anchor: the shoulder



S: shoulder · H: hand · explanatory fields, not measured boundaries

Kandel 6e, Fig. 34–6B, p. 828 · a different bimodal neuron; receptive-field summary; companion geometry illustrative

Parietal cortex · a shoulder-centered workspace

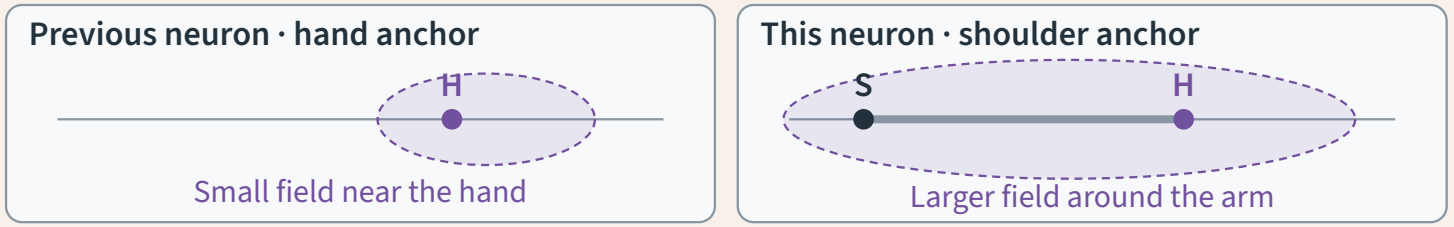


Shoulder-centered example

- Touch field at the shoulder
- Larger nearby visual field

Anchor: the shoulder

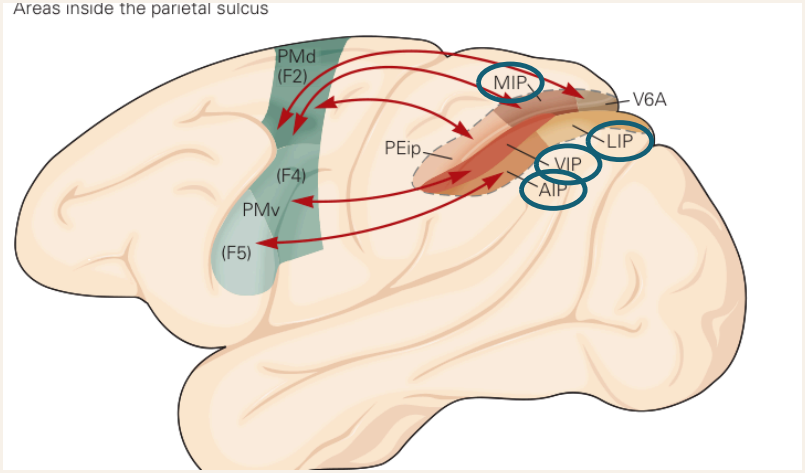
- Learned tool use
- Extended workspace
- Shoulder still the anchor



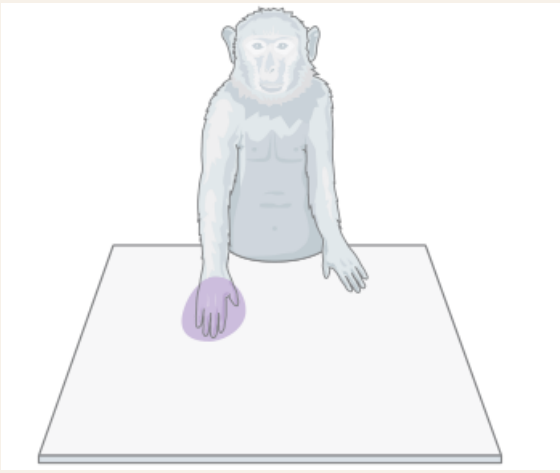
S: shoulder · H: hand · explanatory fields, not measured boundaries

Kandel 6e, Fig. 34–6B, p. 828 · a different bimodal neuron; receptive-field summary; companion geometry illustrative

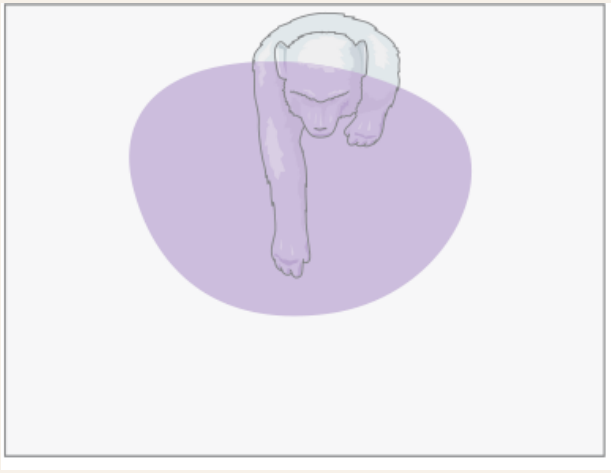
Section B summary: PPC relates body and environment



LIP gaze · VIP head · MIP reach · AIP grasp



Hand-anchored space



Shoulder-anchored space

TAKE HOME PPC relates sensory information to the body for action.

Kandel 6e, Figs. 34–2B (lower), p. 820; 34–6A/B, p. 828 · area experiments in Fig. 34–5

C · What specifies the next action?

Current sensory information ⟷ **Internally maintained plan**



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

C · What specifies the next action?

Current sensory information ⟷ **Internally maintained plan**

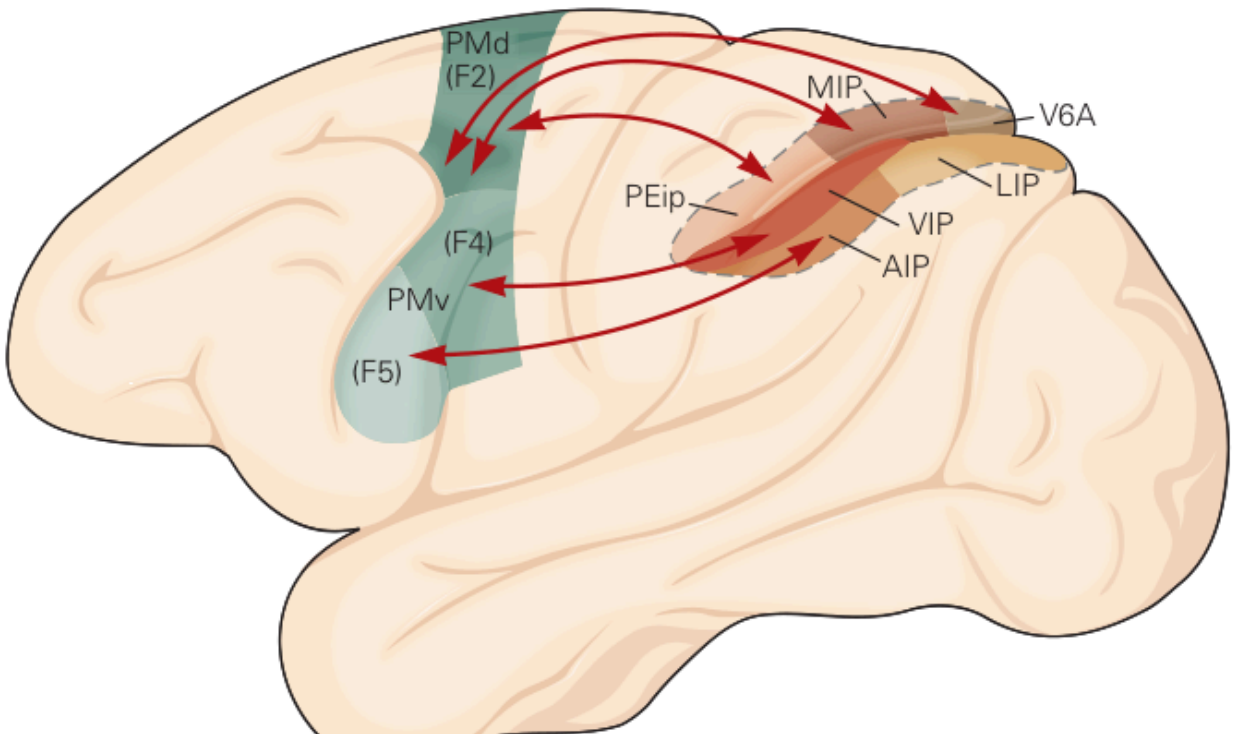


Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

- Actions combine current sensory guidance and maintained plans.
- These are relative emphases, not separate systems.

Locate the connected action circuits

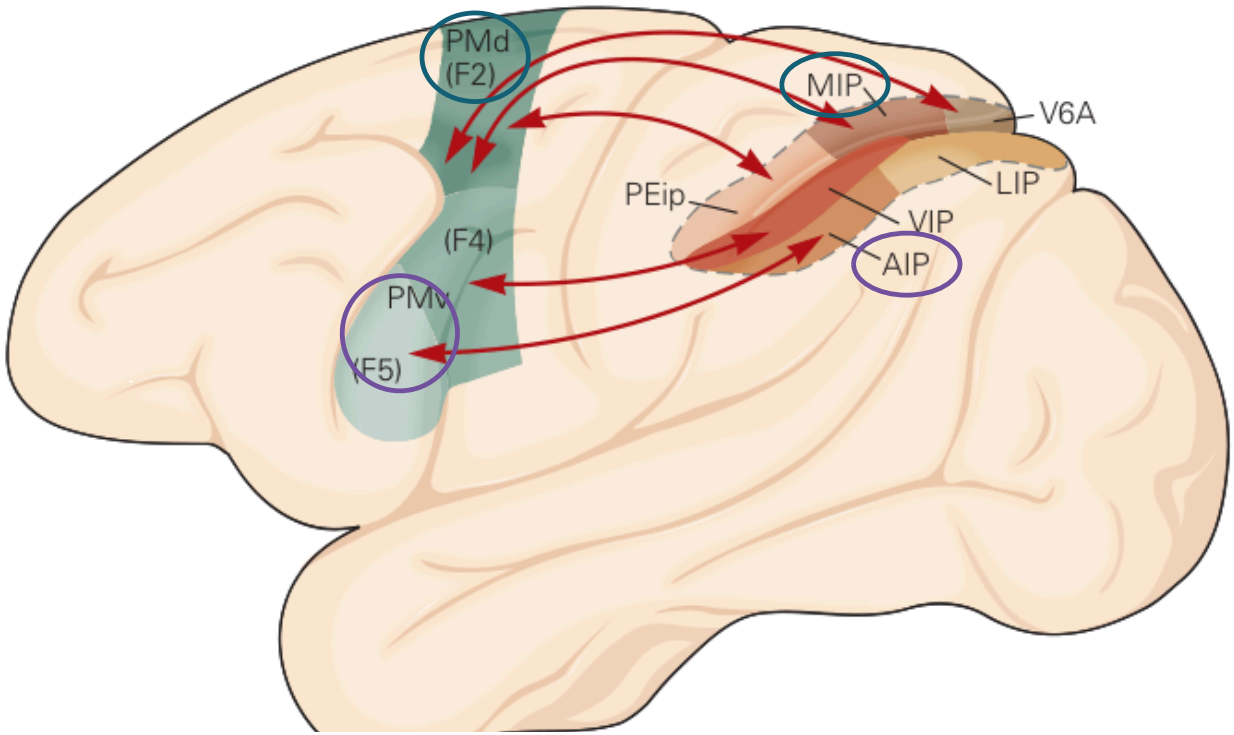
Areas inside the parietal sulcus



Kandel 6e, Fig. 34–2B (lower view), p. 820 · parietofrontal pathways; rings added

Locate the connected action circuits

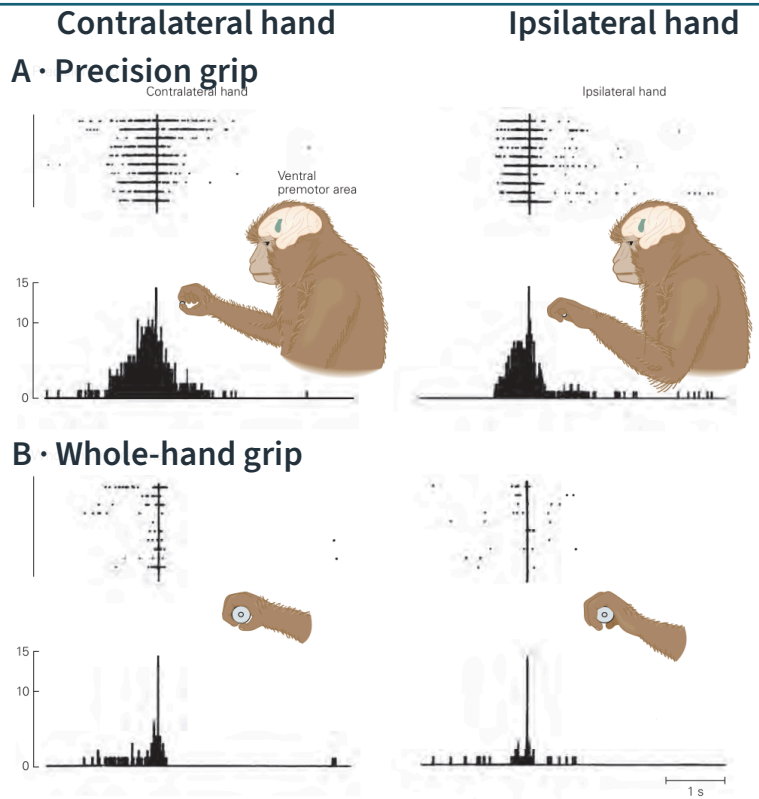
Areas inside the parietal sulcus



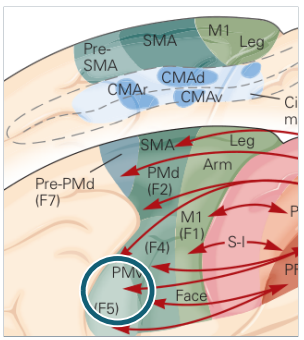
Kandel 6e, Fig. 34–2B (lower view), p. 820 · parietofrontal pathways; rings added

- **Reach branch:** MIP ↔ PMd
- **Grasp branch:** AIP ↔ PMv
- Routes run both ways

PMv · grip form with either hand

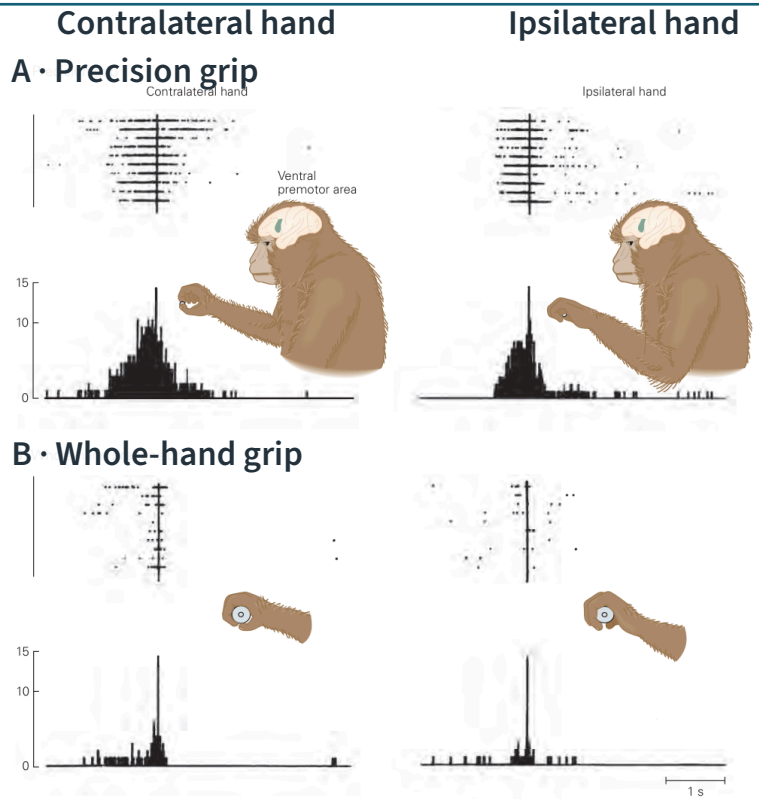


Ventral premotor area

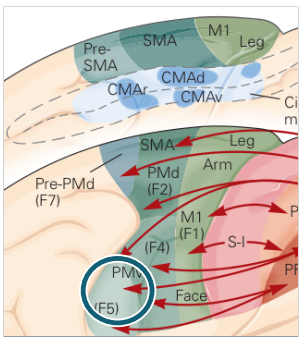


Kandel 6e, Fig. 34–12A/B, p. 837; 34–2B, p. 820 · precision versus whole-hand grip

PMv · grip form with either hand



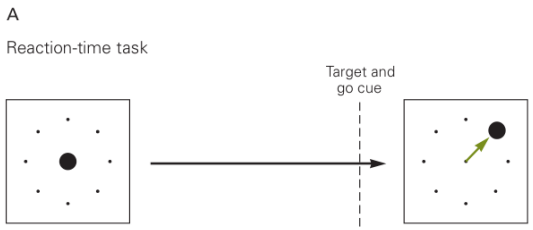
Ventral premotor area



- This cell: precision, both hands
- Other cells prefer other grips
- PMv cells are grip selective

Kandel 6e, Fig. 34–12A/B, p. 837; 34–2B, p. 820 · precision versus whole-hand grip

PMd · preparation before movement

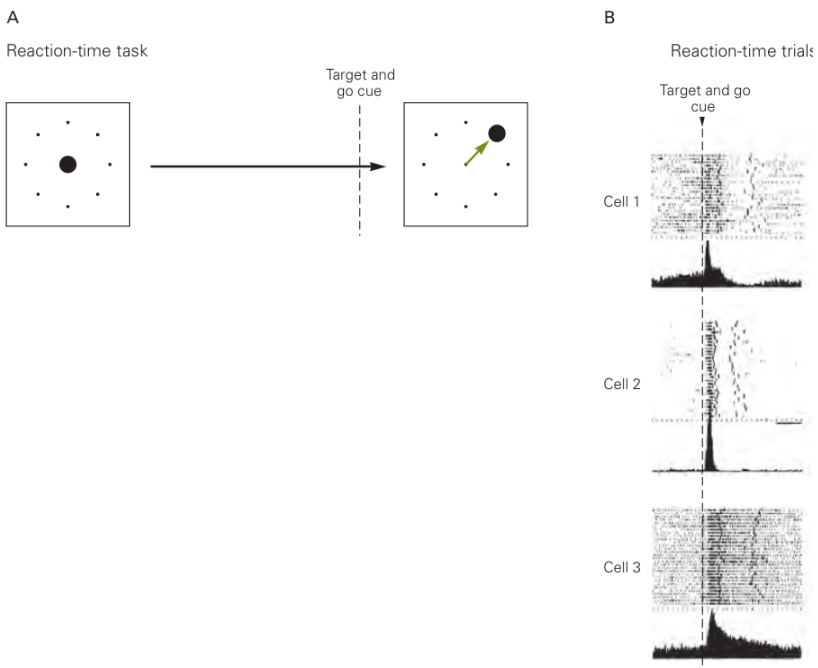


Dorsal premotor cortex

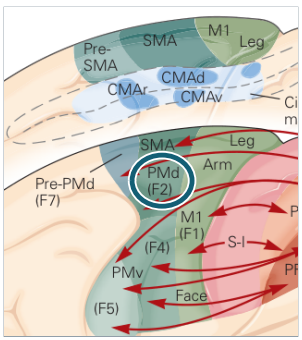
- Reaction time: target = go cue

Kandel 6e, Fig. 34–4A/B, p. 823; 34–2B, p. 820 · three premotor neurons, recorded in PMd (Crammond & Kalaska, 2000)

PMd · preparation before movement



Dorsal premotor cortex

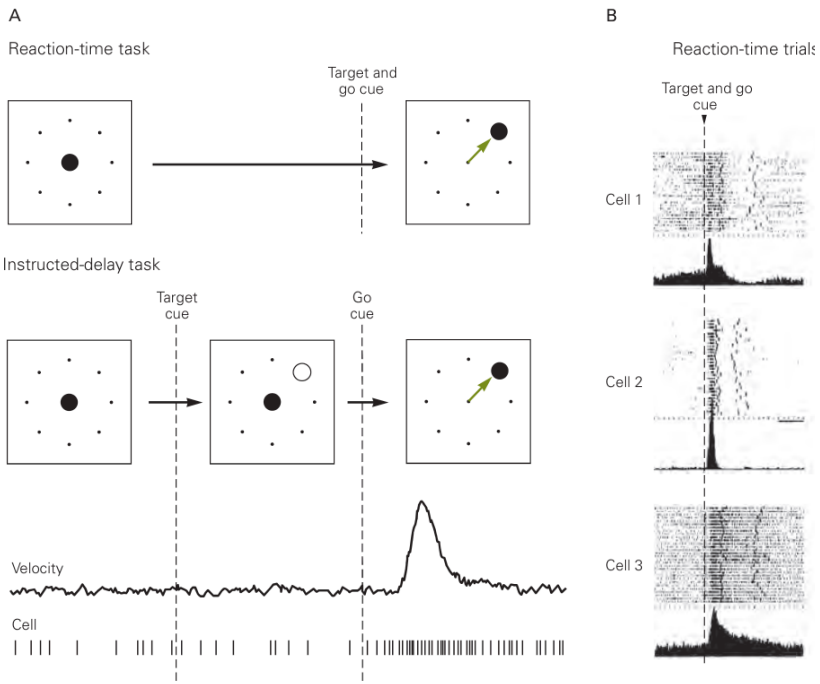


- Reaction time: target = go cue

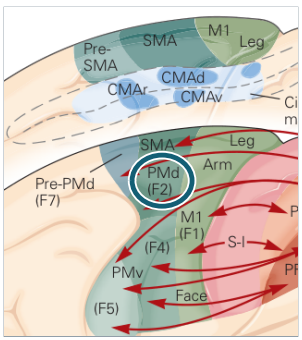
> - Fires after the cue

Kandel 6e, Fig. 34–4A/B, p. 823; 34–2B, p. 820 · three premotor neurons, recorded in PMd (Crammond & Kalaska, 2000)

PMd · preparation before movement



Dorsal premotor cortex

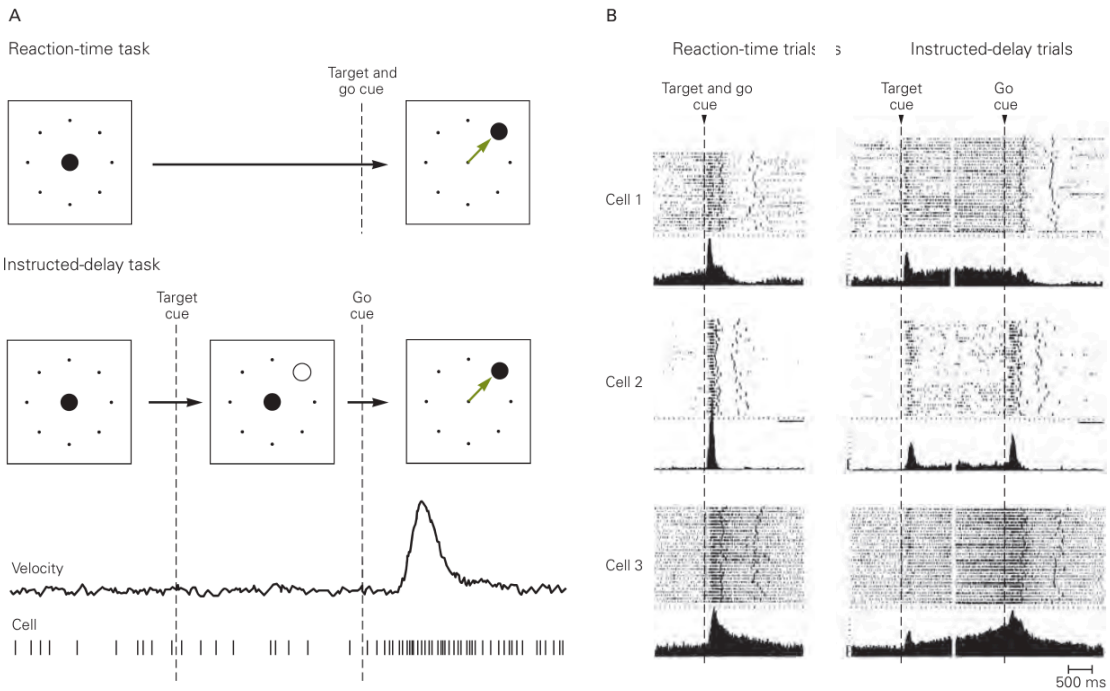


- Reaction time: target = go cue
- Instructed delay: target, then go

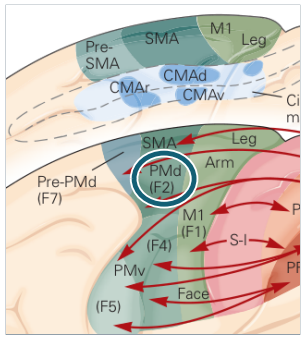
> - Fires after the cue

Kandel 6e, Fig. 34–4A/B, p. 823; 34–2B, p. 820 · three premotor neurons, recorded in PMd (Crammond & Kalaska, 2000)

PMd · preparation before movement



Dorsal premotor cortex



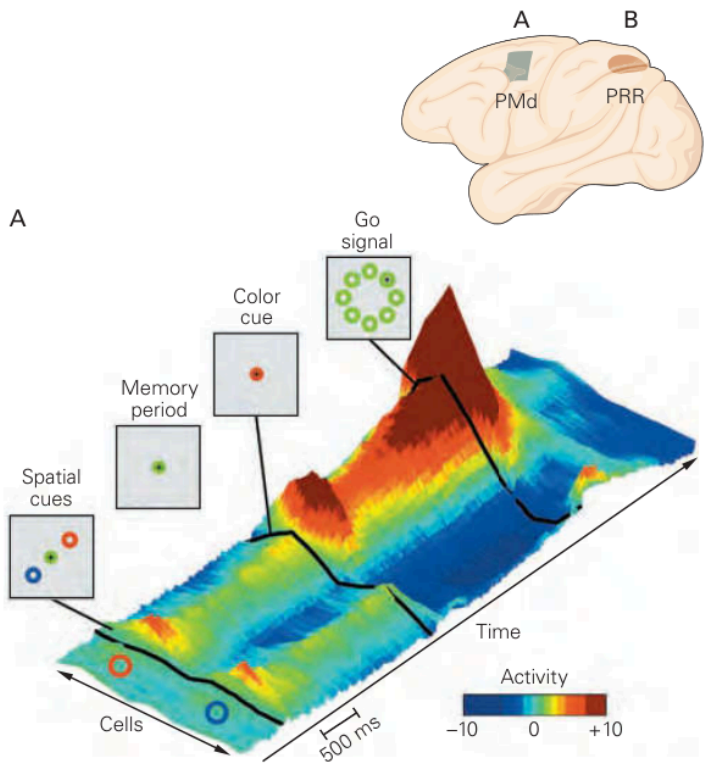
- Reaction time: target = go cue
- Instructed delay: target, then go

> - Fires after the cue

> - Delay: tuned activity before go
> - Planning and movement mix
> - Cue → maintained reach plan

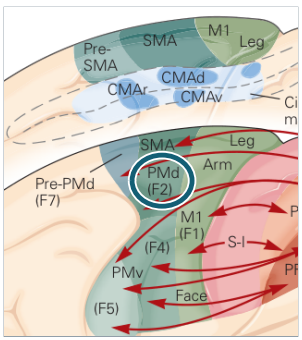
Kandel 6e, Fig. 34–4A/B, p. 823; 34–2B, p. 820 · three premotor neurons, recorded in PMd (Crammond & Kalaska, 2000)

PMd · potential reaches before a choice



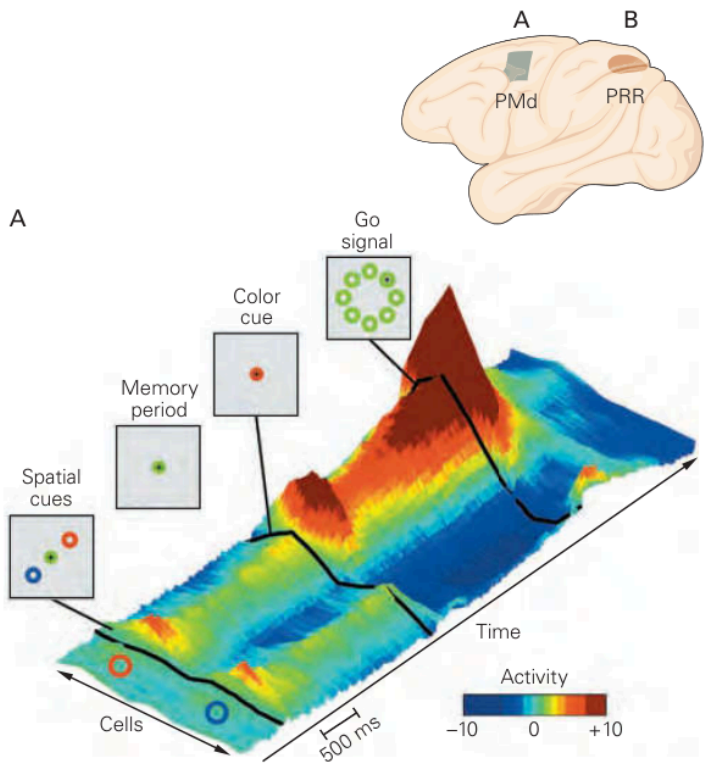
Kandel 6e, Fig. 34–8A, p. 832; 34–2B, p. 820 · two target cues, color choice cue and go cue

Dorsal premotor cortex



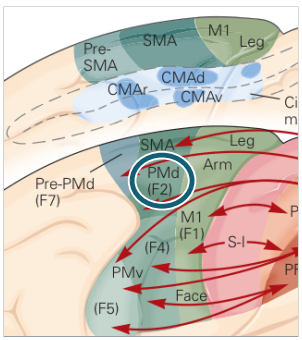
- Two potential targets: red and blue
- A color cue chooses one
- Go signal: reach

PMd · potential reaches before a choice



Kandel 6e, Fig. 34–8A, p. 832; 34–2B, p. 820 · two target cues, color choice cue and go cue

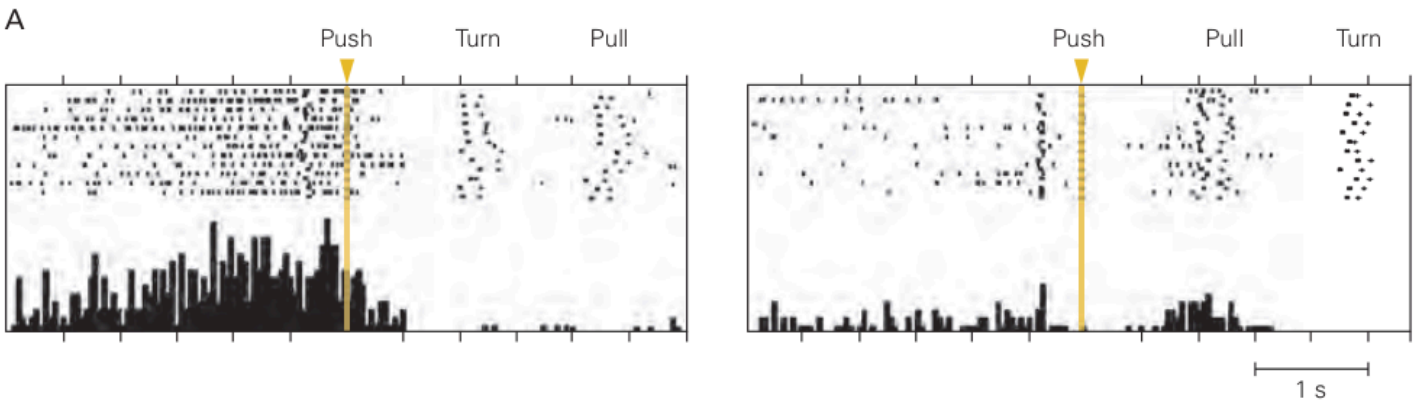
Dorsal premotor cortex



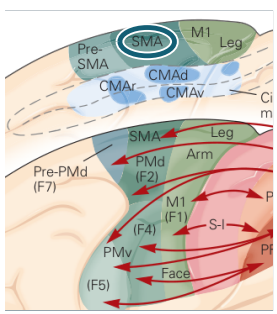
- Two potential targets: red and blue
- A color cue chooses one
- Go signal: reach

> - Delay: both reaches present
> - After color cue: chosen reach

SMA · preparation depends on the remembered sequence



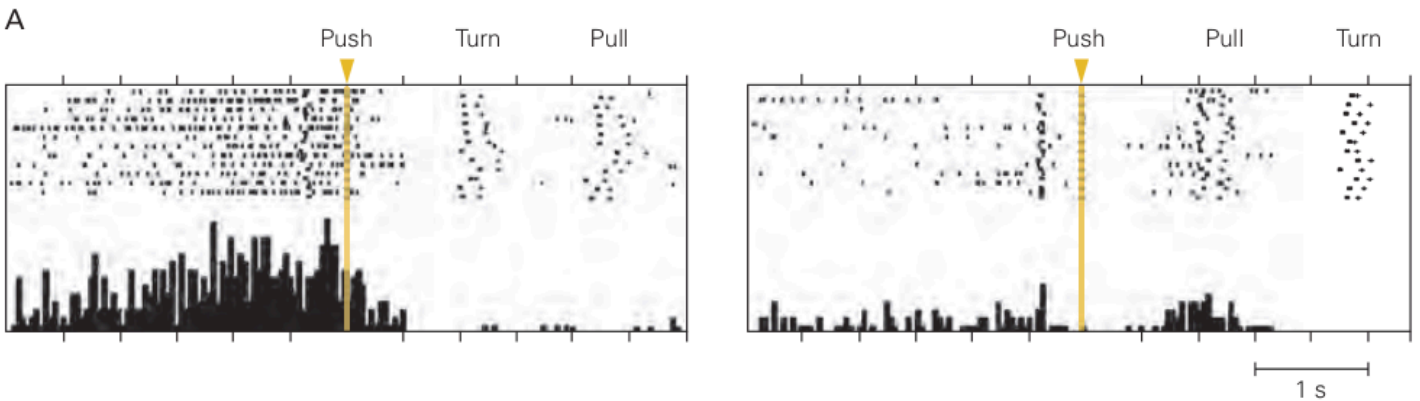
Supplementary motor area



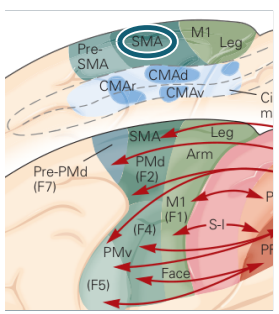
- Memorized sequences
- Both begin with a push

Kandel 6e, Fig. 34–7A, p. 830; 34–2B, p. 820 · sequence comparison; Kandel's category is the supplementary motor complex

SMA · preparation depends on the remembered sequence



Supplementary motor area

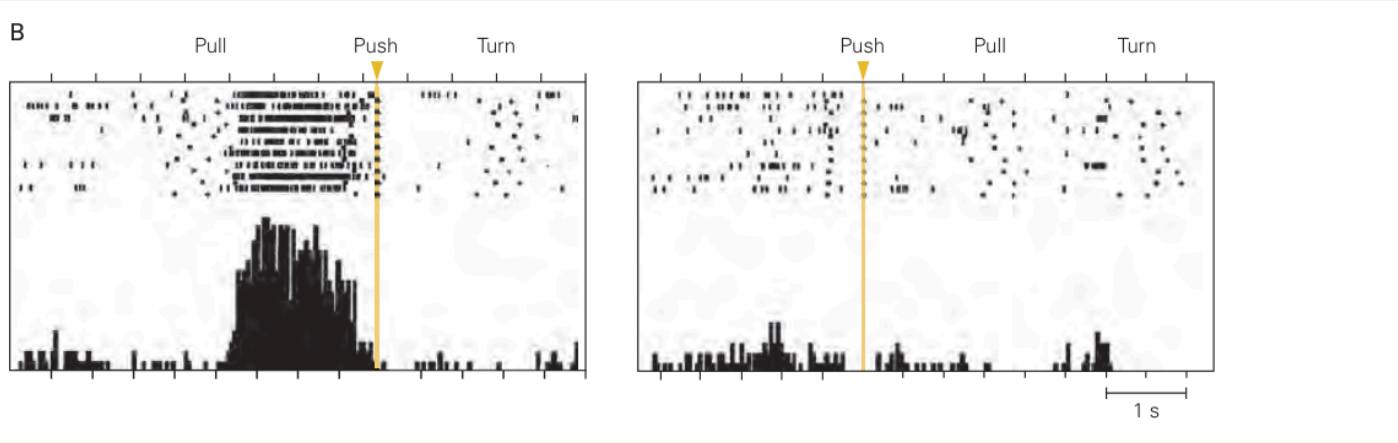


- Memorized sequences
- Both begin with a push

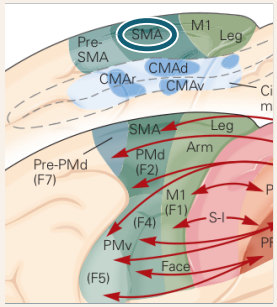
- Push–turn–pull: active before push
- Push–pull–turn: little activity
- Order, not the first act

Kandel 6e, Fig. 34–7A, p. 830; 34–2B, p. 820 · sequence comparison; Kandel's category is the supplementary motor complex

SMA · activity at a transition



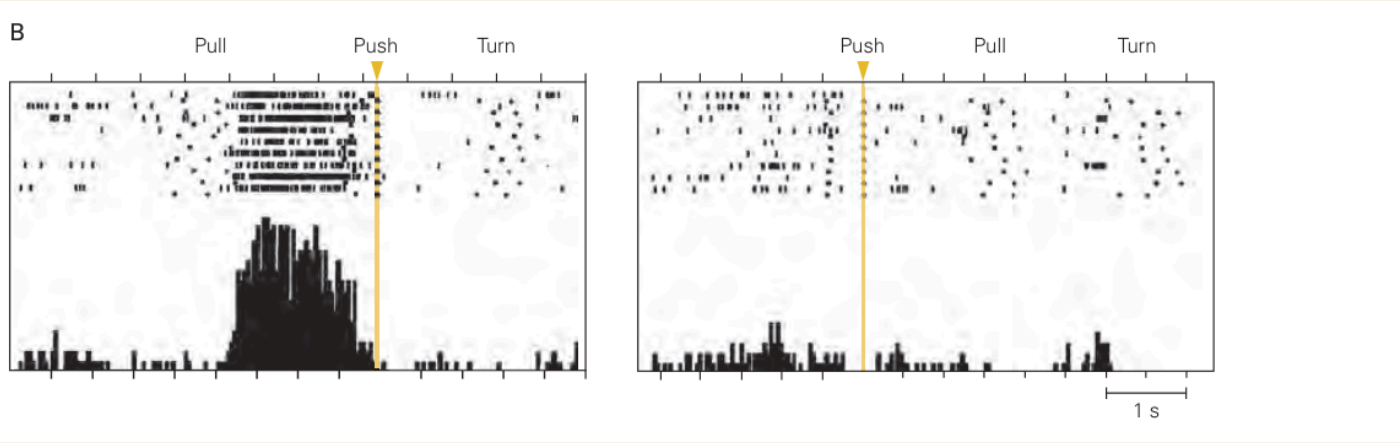
Supplementary motor area



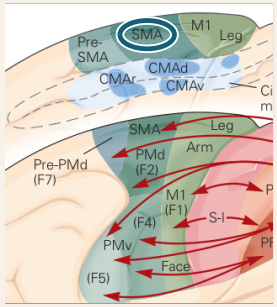
- Left: pull, then push
- Right: other orders

Kandel 6e, Fig. 34–7B, p. 830; 34–2B, p. 820 · sequence transition; Kandel's category is the supplementary motor complex

SMA · activity at a transition



Supplementary motor area

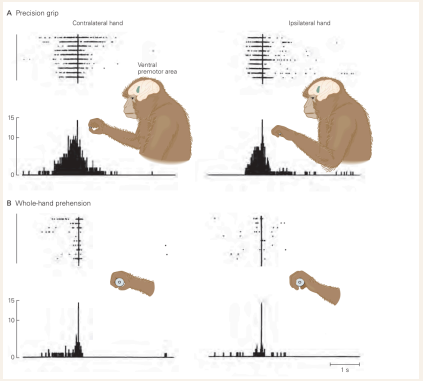


- Left: pull, then push
- Right: other orders

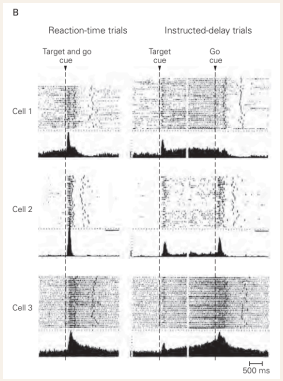
- Active between pull and push
- Not for push first or pull then turn
- Order organizes the next act

Kandel 6e, Fig. 34–7B, p. 830; 34–2B, p. 820 · sequence transition; Kandel's category is the supplementary motor complex

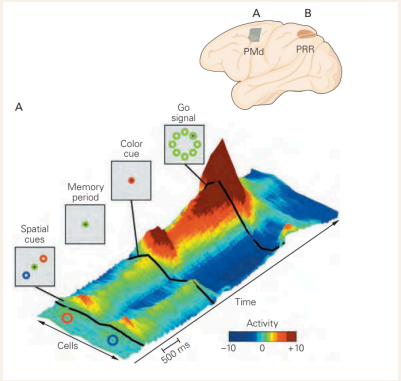
Section C summary: premotor cortex organizes the action



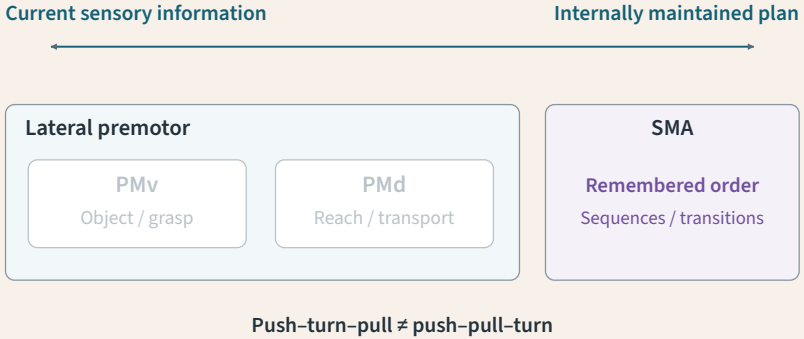
PMv · grip form



PMd · preparation



PMd · potential reaches

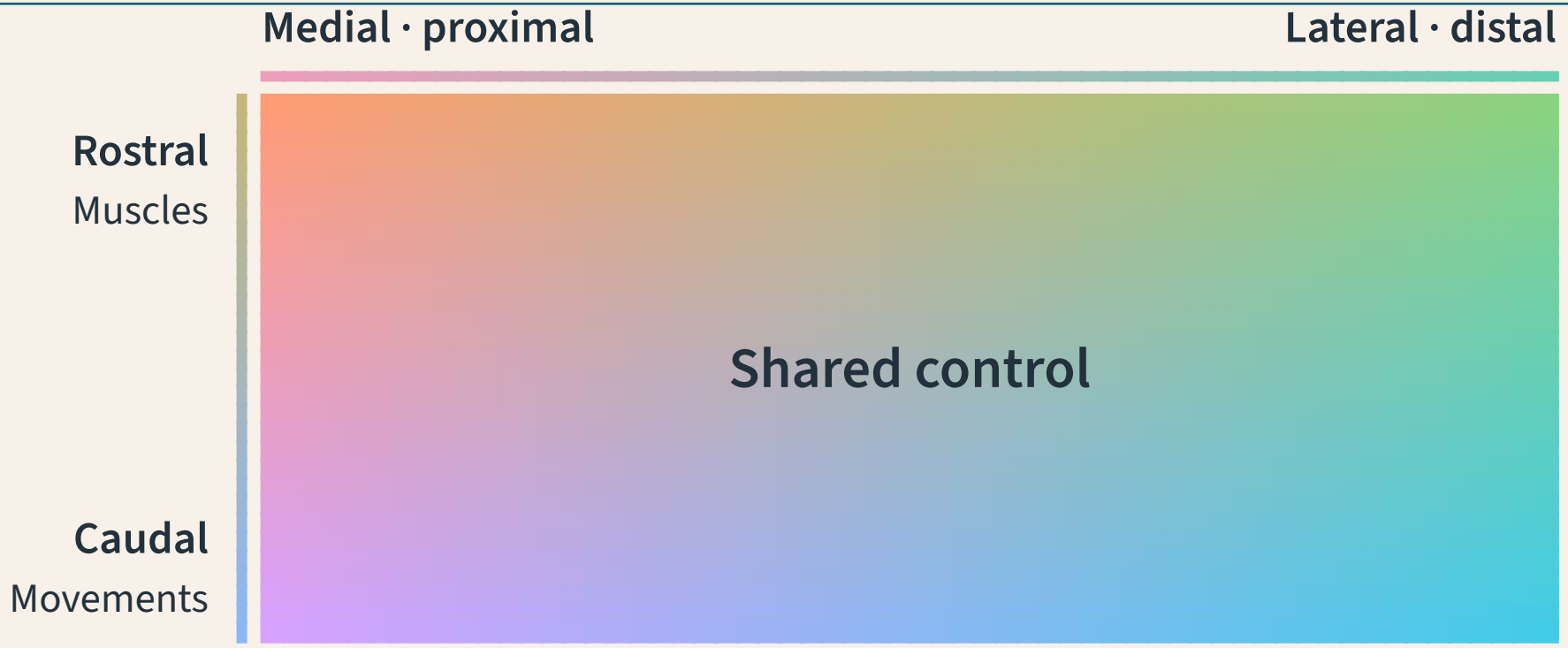


SMA · remembered order

TAKE HOME Premotor activity organizes actions before and during movement.

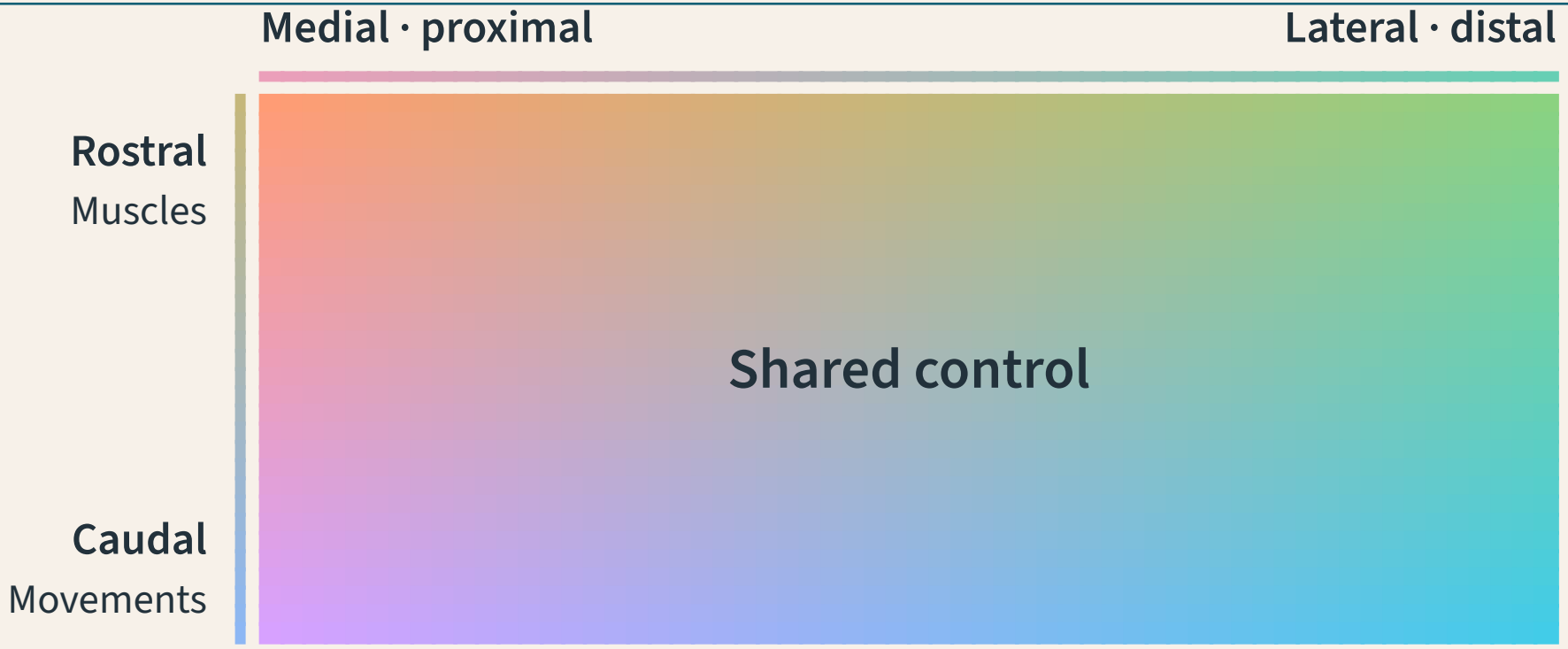
Kandel 6e, Figs. 34–12, 34–4B, 34–8A · sequence schematic after Fig. 34–7

D · M1: overlapping control



Kandel 6e, Ch. 34; Kolasinski et al. (2020) · conceptual sketch, not a measured cortical atlas

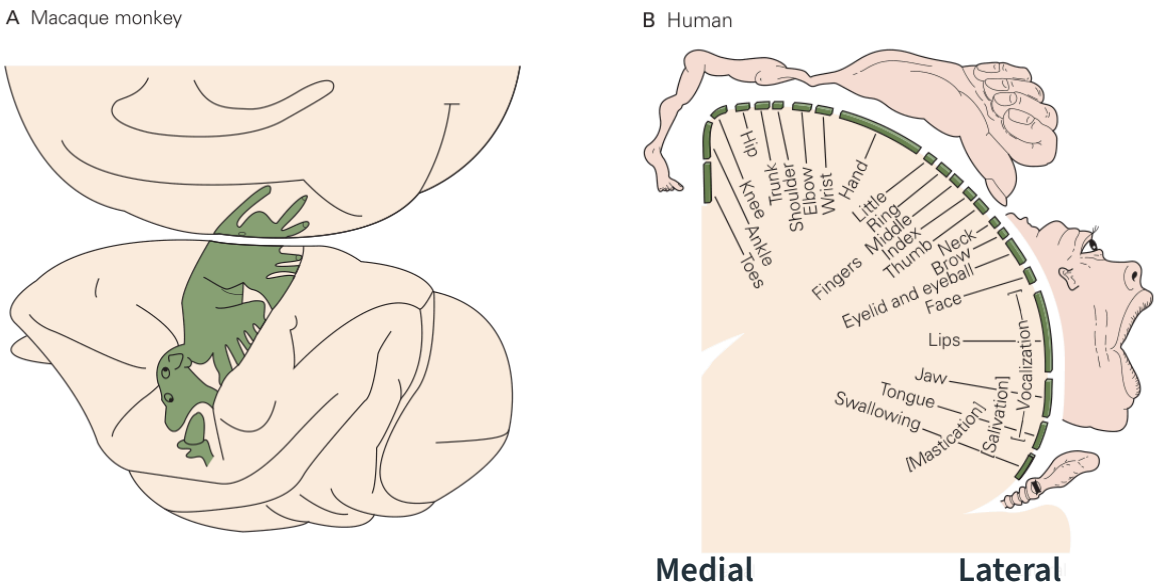
D · M1: overlapping control



Kandel 6e, Ch. 34; Kolasinski et al. (2020) · conceptual sketch, not a measured cortical atlas

Two overlapping dimensions: which effector, and muscle versus movement emphasis.

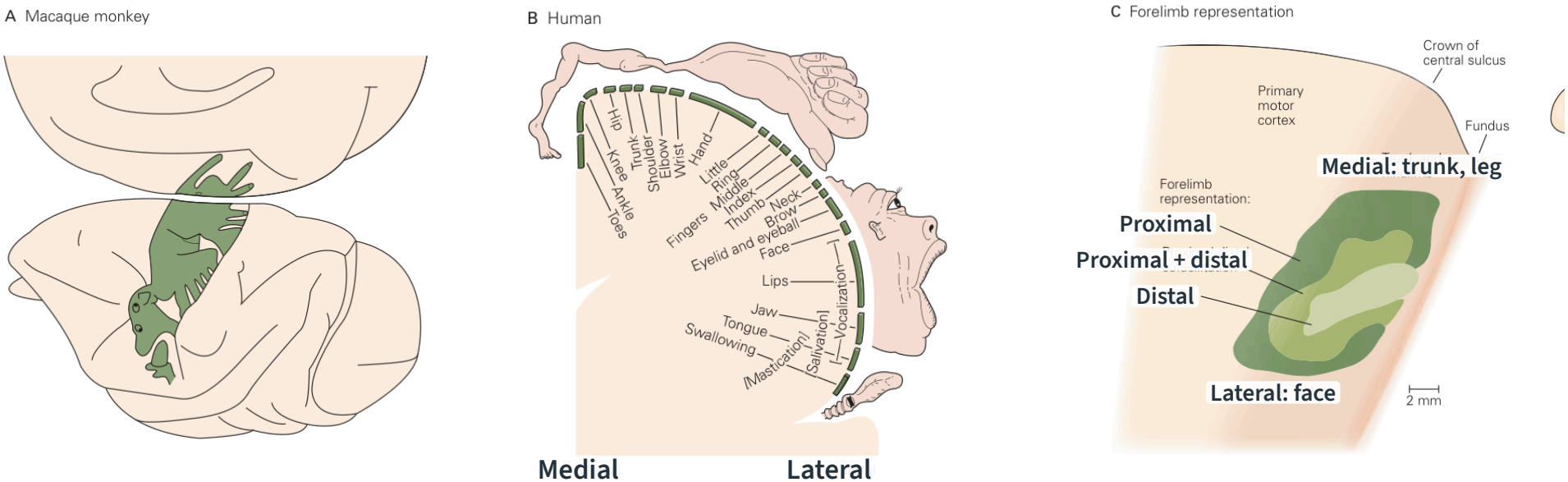
M1 map · broad order with extensive overlap



- Monkey and human: leg medial, face lateral
- Hand and mouth areas are large

Kandel 6e, Fig. 34–16A–C, p. 842 · stimulation-based maps; A macaque, B human, C macaque arm (Park et al., 2001)

M1 map · broad order with extensive overlap

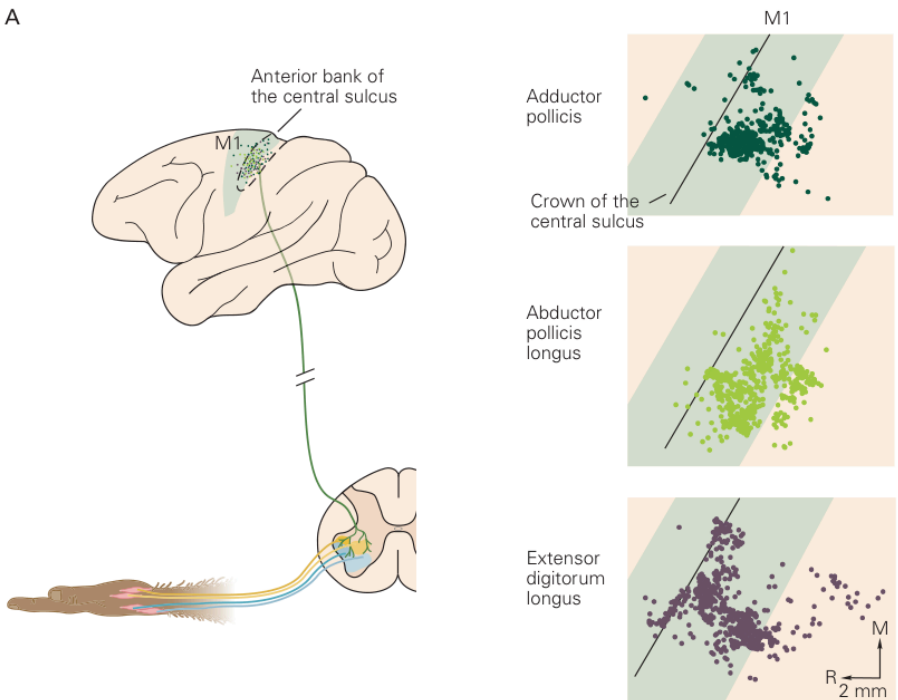


- Monkey and human: leg medial, face lateral
- Hand and mouth areas are large

- Distal arm: central core
- Proximal arm: surrounding ring
- Wide overlap between them

Kandel 6e, Fig. 34–16A–C, p. 842 · stimulation-based maps; A macaque, B human, C macaque arm (Park et al., 2001)

CM cells · overlapping cortical distributions

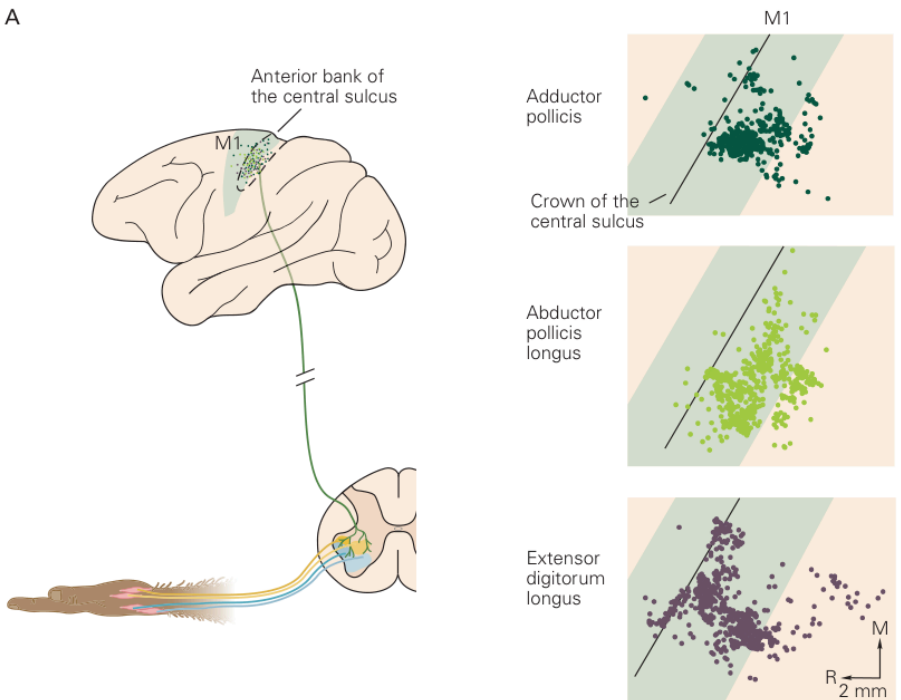


Kandel 6e, Fig. 34–17A, p. 843 · CM-cell distributions

Corticomotoneuronal cells

- Direct projection to spinal motor neurons
- Cells for three hand muscles mapped

CM cells · overlapping cortical distributions



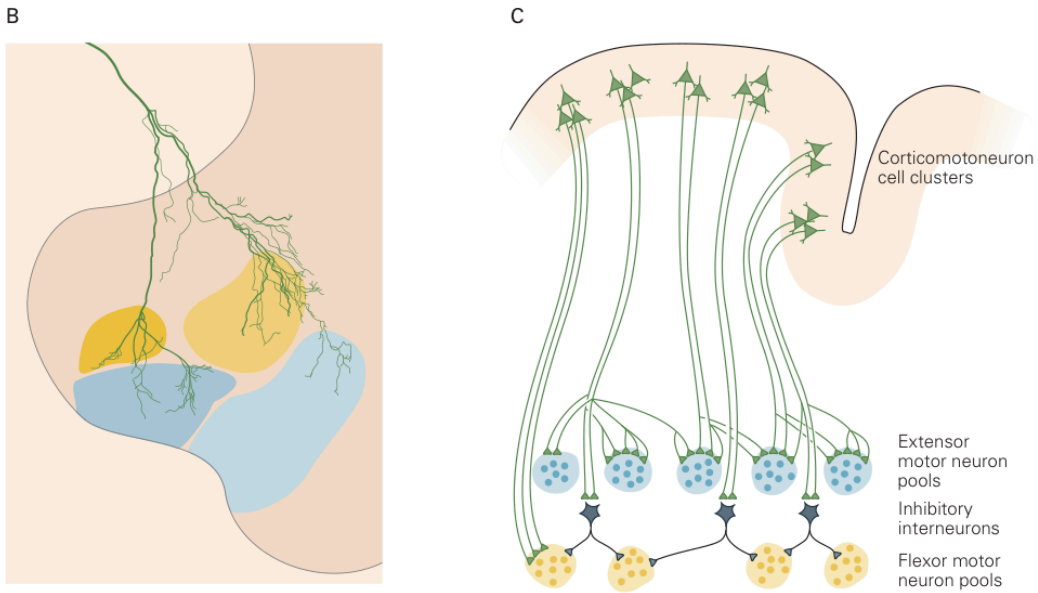
Kandel 6e, Fig. 34–17A, p. 843 · CM-cell distributions

Corticomotoneuronal cells

- Direct projection to spinal motor neurons
- Cells for three hand muscles mapped

> - Caudal M1, anterior bank
> - Widely distributed
> - Overlapping across muscles

CM output · combinations of muscles

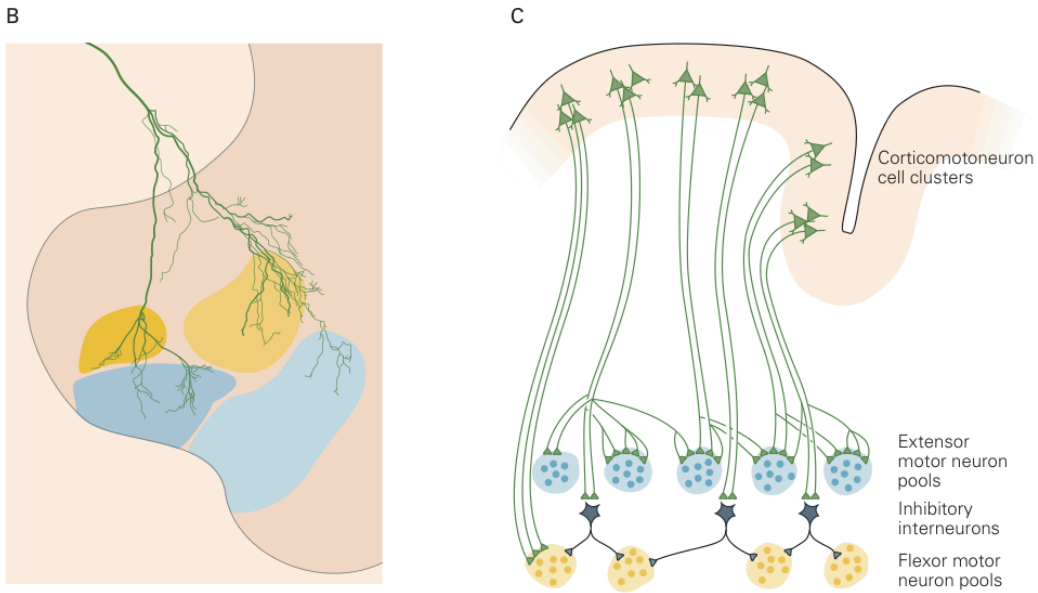


Corticomotoneuronal output

- B: one CM axon in the ventral horn
- C: direct and interneuron routes

Kandel 6e, Fig. 34–17B/C, p. 843 · divergent terminals and spinal routes

CM output · combinations of muscles



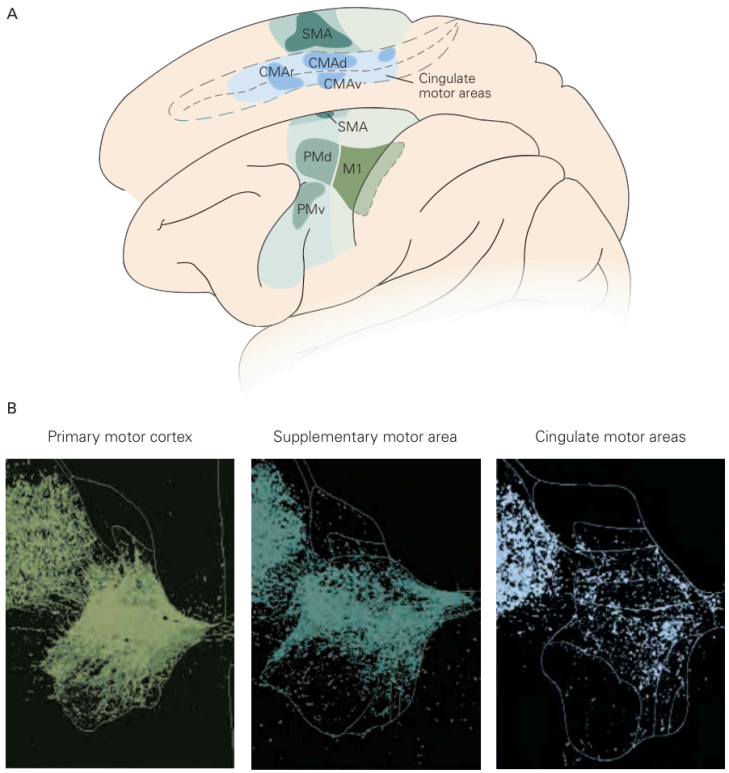
Corticomotoneuronal output

- B: one CM axon in the ventral horn
- C: direct and interneuron routes

- One axon reaches several motor pools
- Outputs combine muscles

Kandel 6e, Fig. 34–17B/C, p. 843 · divergent terminals and spinal routes

Descending control · M1 among several sources

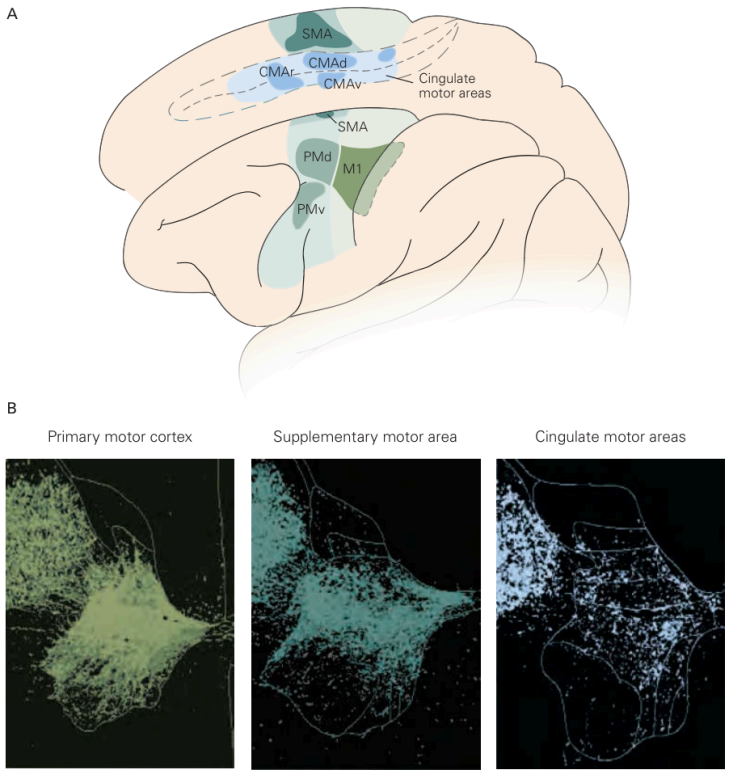


Kandel 6e, Fig. 34–3A/B, p. 822 · cortical origins and spinal terminations

Corticospinal origins

- A: arm and hand origins
- B: spinal terminations

Descending control · M1 among several sources



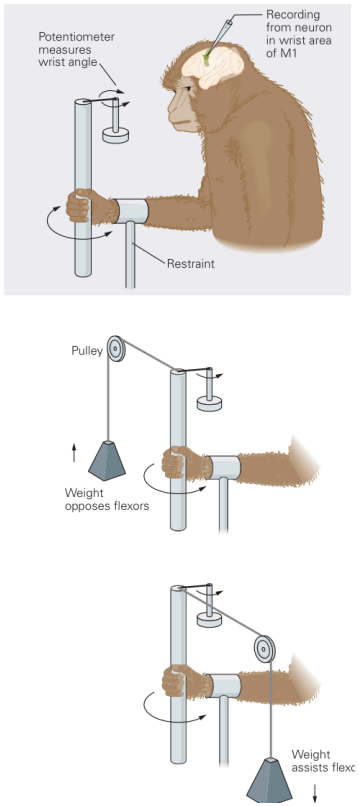
Kandel 6e, Fig. 34–3A/B, p. 822 · cortical origins and spinal terminations

Corticospinal origins

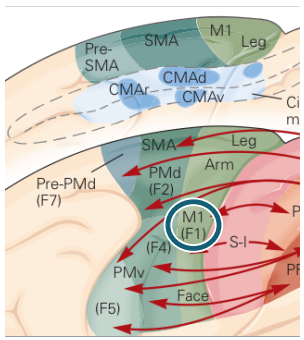
- A: arm and hand origins
- B: spinal terminations

> - Many frontal areas project
> - M1 is one of several sources

M1 · same displacement, different muscle output



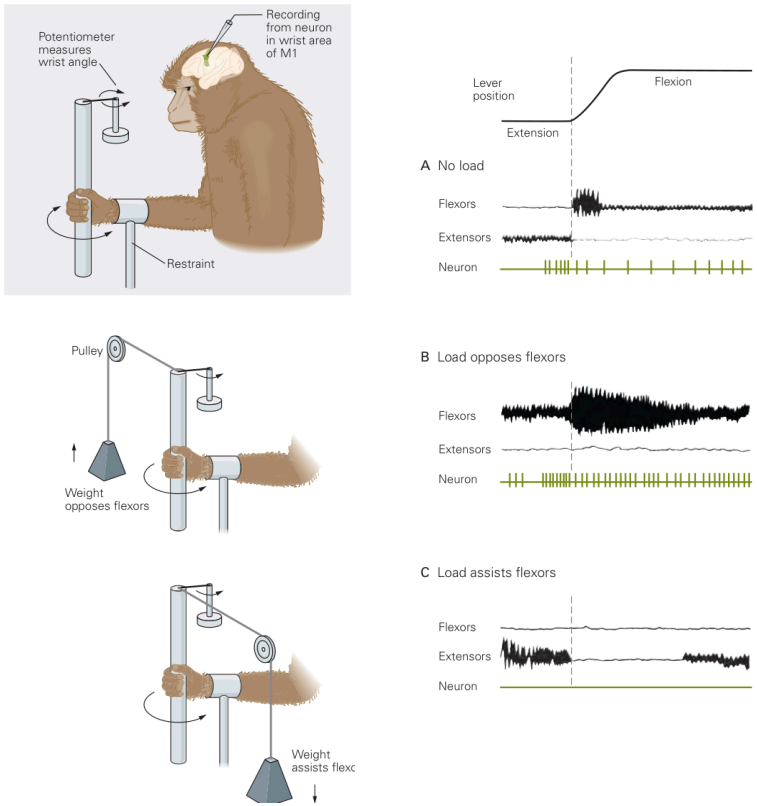
Primary motor cortex, wrist area



- Same wrist flexion each time
- A: no load · B: load opposes flexors · C: load assists

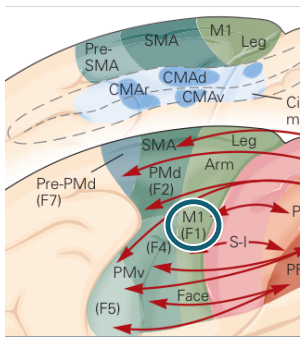
Kandel 6e, Fig. 34–18A–C, p. 846; 34–2B, p. 820 · no load, opposing load and assisting load

M1 · same displacement, different muscle output



Kandel 6e, Fig. 34–18A–C, p. 846; 34–2B, p. 820 · no load, opposing load and assisting load

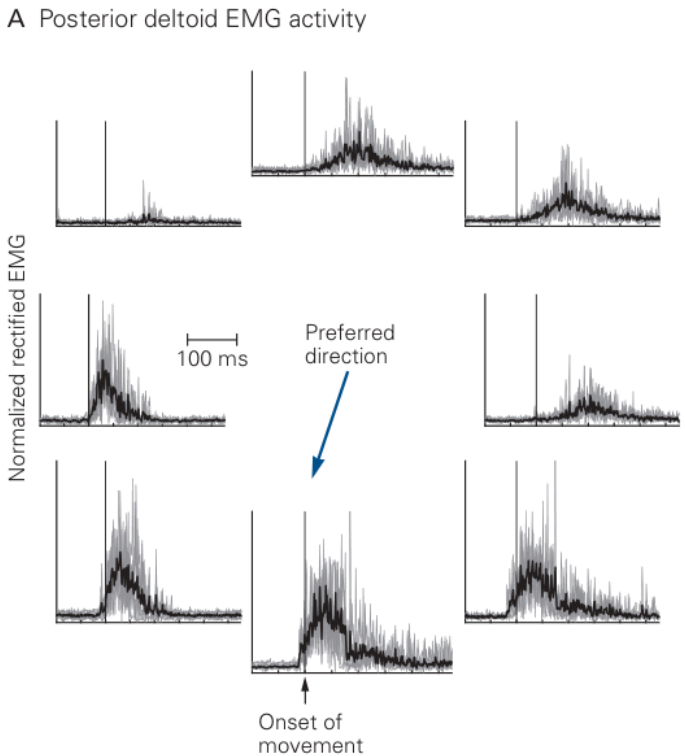
Primary motor cortex, wrist area



- Same wrist flexion each time
- A: no load · B: load opposes flexors · C: load assists

- Flexor EMG follows the load
- So does the neuron
- Activity tracks force, not displacement

Muscle · broad tuning to reach direction

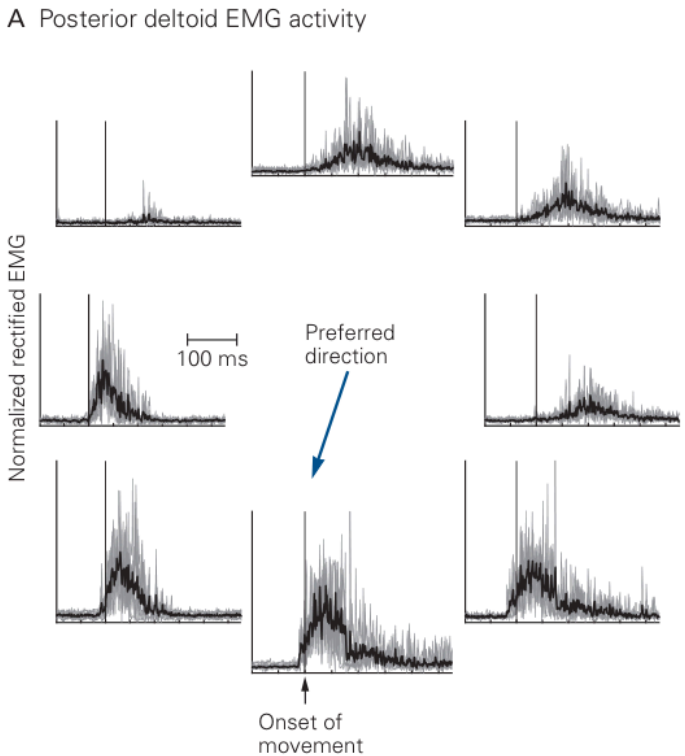


Posterior deltoid

- Reaches in eight directions
- EMG aligned on movement onset

Kandel 6e, Fig. 34–20A, p. 848 · posterior deltoid EMG

Muscle · broad tuning to reach direction



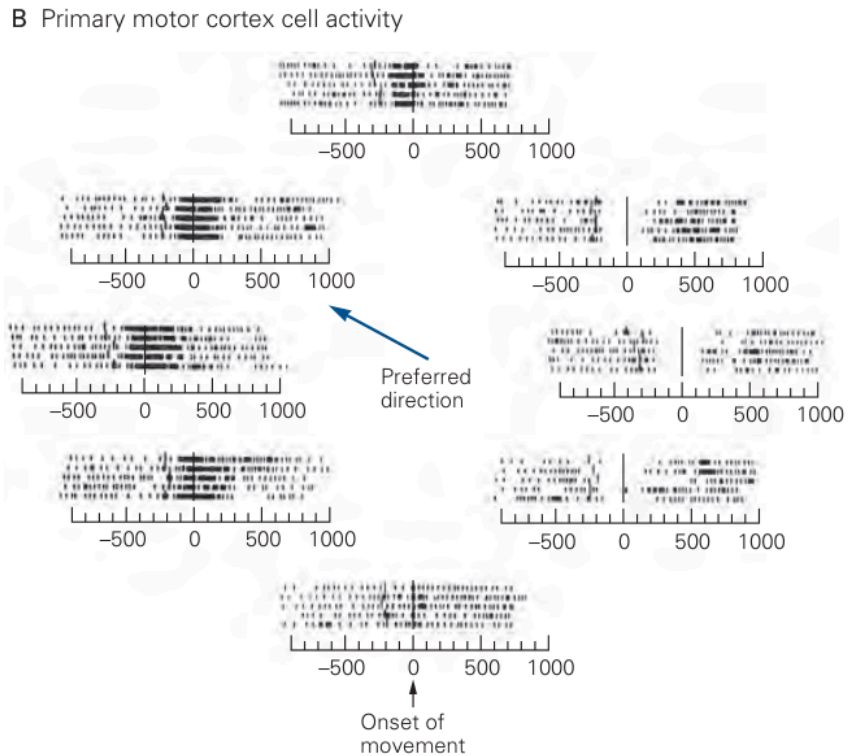
Posterior deltoid

- Reaches in eight directions
- EMG aligned on movement onset

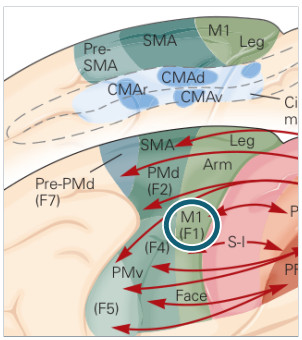
- Peak near 250°
- Also active nearby

Kandel 6e, Fig. 34–20A, p. 848 · posterior deltoid EMG

M1 neuron · broad tuning to reach direction



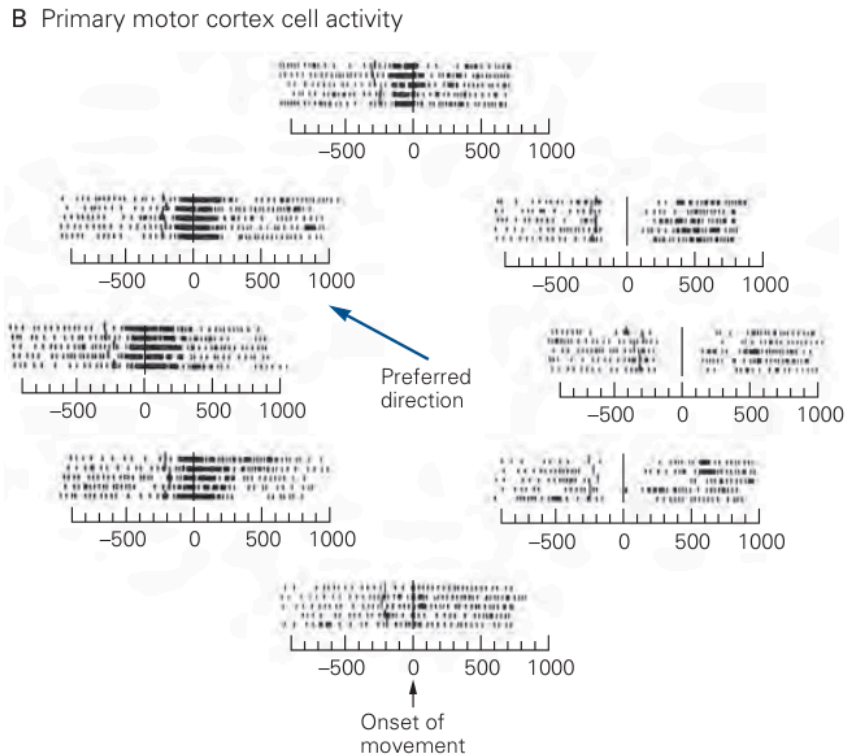
Primary motor cortex



- The same eight reach directions

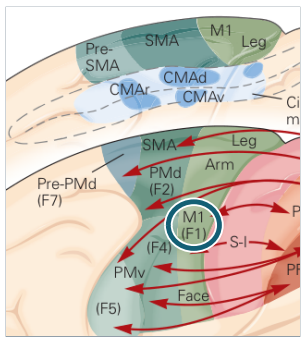
Kandel 6e, Fig. 34–20B, p. 848; 34–2B, p. 820 · example M1 neuron

M1 neuron · broad tuning to reach direction



Kandel 6e, Fig. 34–20B, p. 848; 34–2B, p. 820 · example M1 neuron

Primary motor cortex

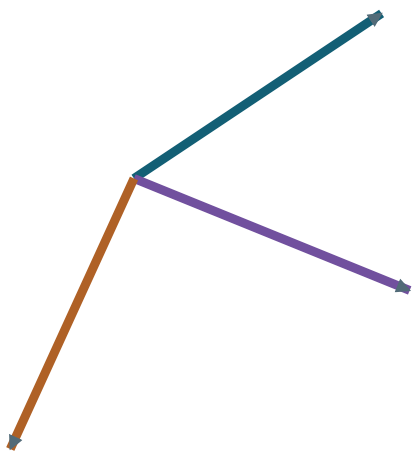


- The same eight reach directions

- Maximal near 135–180°
- Lower across other directions
- Lowest opposite the preferred

Weight preferred directions, then add the vectors

Preferred directions

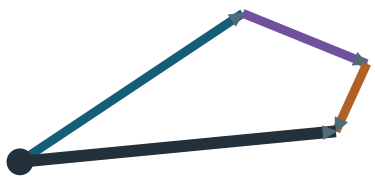


Neuron 1

Neuron 2

Neuron 3

Activity-weighted sum

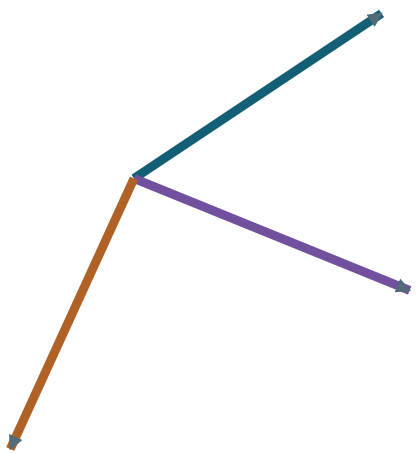


Resultant

Kandel 6e, pp. 849–850 · illustrative three-neuron weighted sum, not recorded values

Weight preferred directions, then add the vectors

Preferred directions

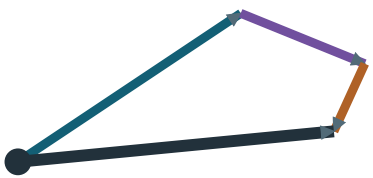


Neuron 1

Neuron 2

Neuron 3

Activity-weighted sum

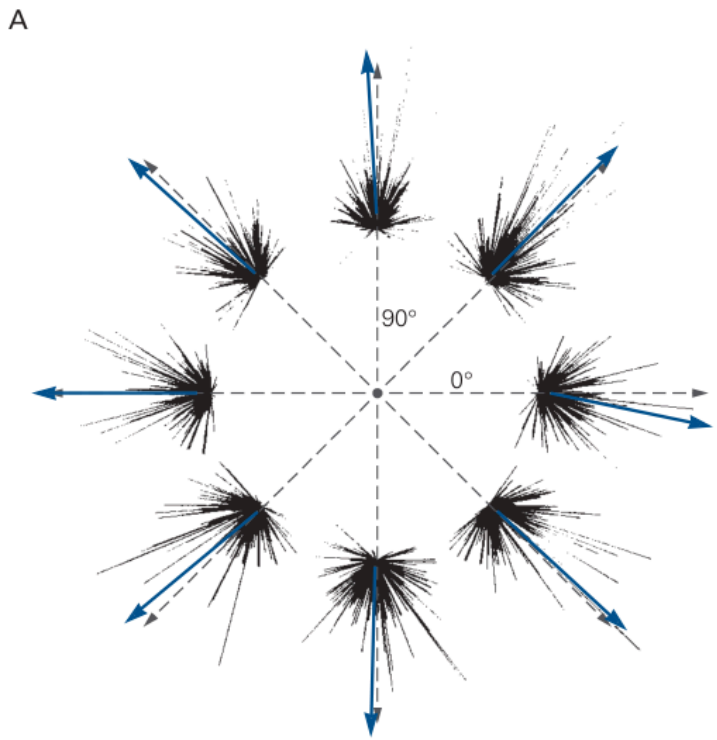


Resultant

Kandel 6e, pp. 849–850 · illustrative three-neuron weighted sum, not recorded values

- Each vector: a preferred direction, scaled by firing
- Their sum: the population vector

Population vector · same cells, different resultants

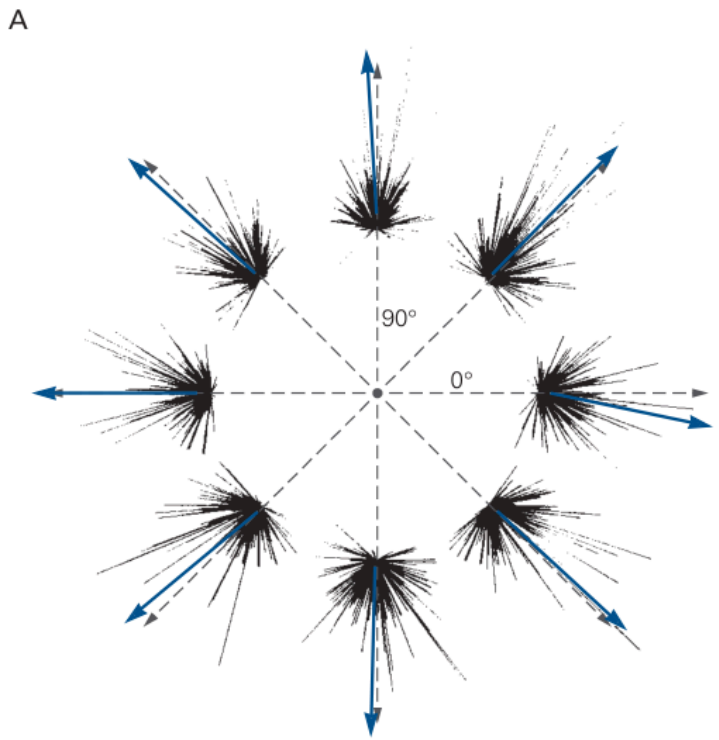


M1 population

- Thin lines: single-cell vectors
- Blue: population vector
- Dashed: reach direction

Kandel 6e, Fig. 34–22A, p. 850 · thin cell vectors, blue population vectors and dashed reach directions

Population vector · same cells, different resultants



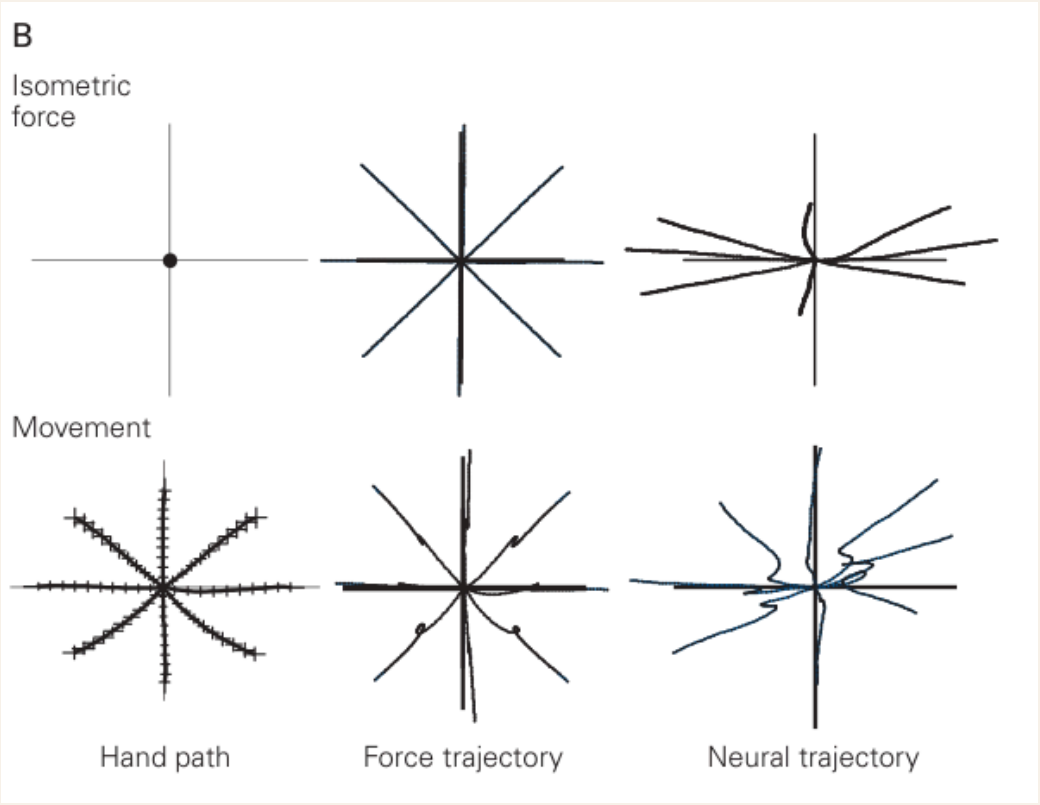
M1 population

- Thin lines: single-cell vectors
- Blue: population vector
- Dashed: reach direction

> - Vector ≈ reach direction
> - Same cells, different weights

Kandel 6e, Fig. 34–22A, p. 850 · thin cell vectors, blue population vectors and dashed reach directions

Population vector · not a pure hand code

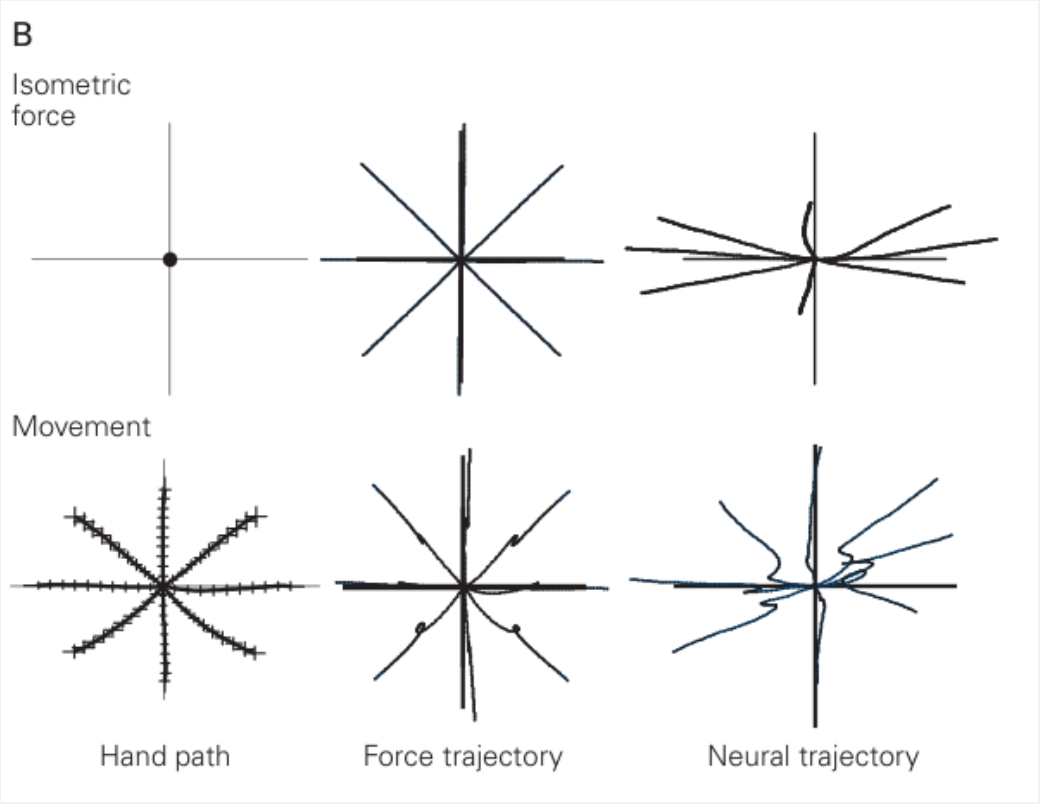


Kandel 6e, Fig. 34–22B, p. 850 · hand path, force trajectory and neural trajectory

M1 population

- Top: isometric force, no movement
- Bottom: moving a heavy handle
- Compare hand, force and neural columns

Population vector · not a pure hand code



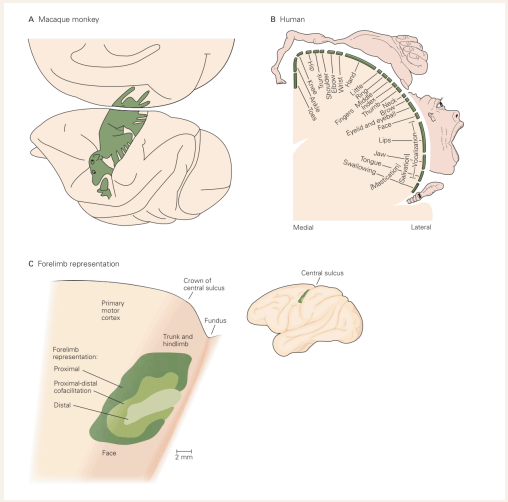
Kandel 6e, Fig. 34–22B, p. 850 · hand path, force trajectory and neural trajectory

M1 population

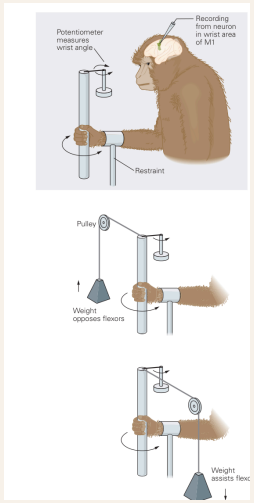
- Top: isometric force, no movement
- Bottom: moving a heavy handle
- Compare hand, force and neural columns

> - Varies with no hand motion
> - They reverse while braking
> - Reflects task mechanics

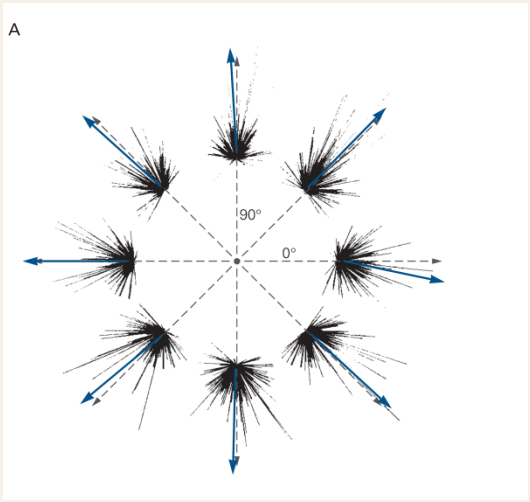
Section D summary: M1 coordinates output for the body and task



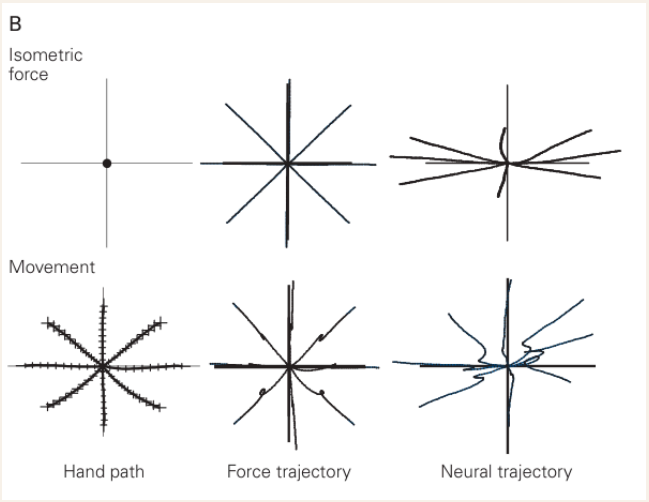
Overlapping map



Force demands



Population vector

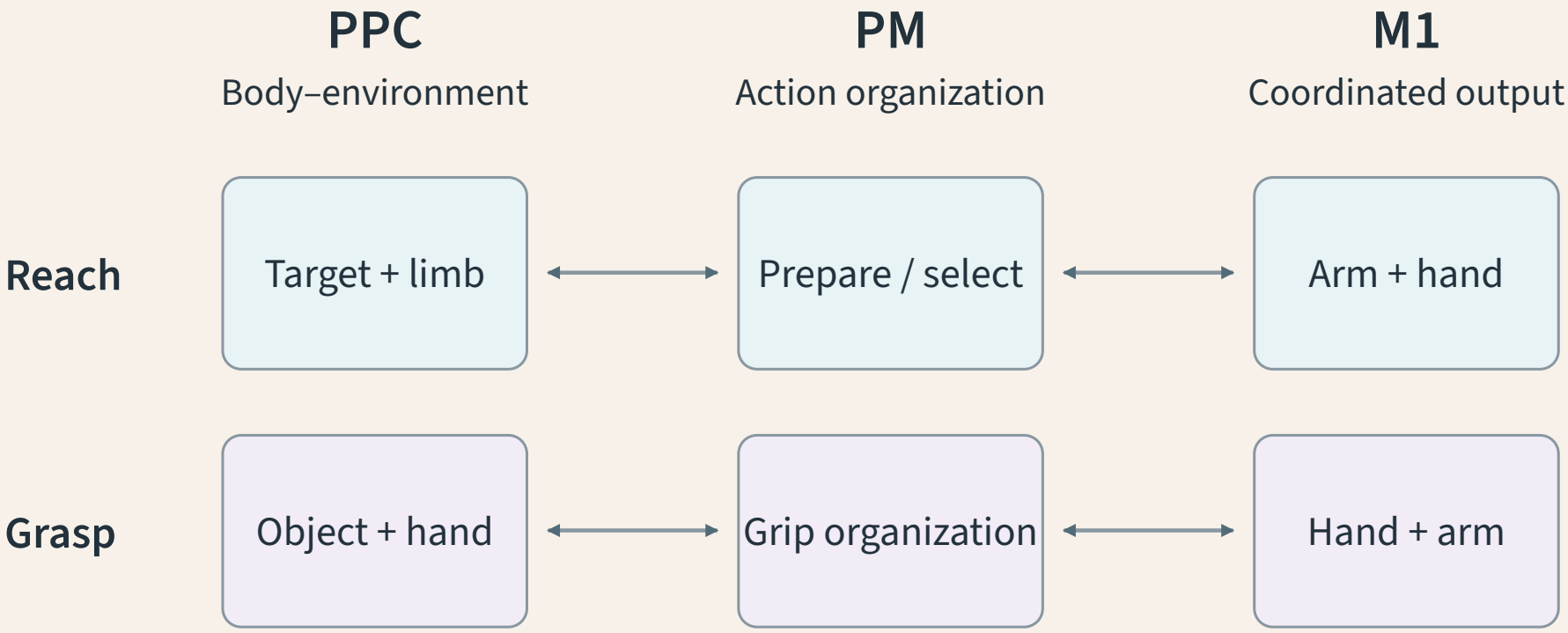


Not a pure hand code

TAKE HOME M1 control is overlapping, distributed and shaped by force demands.

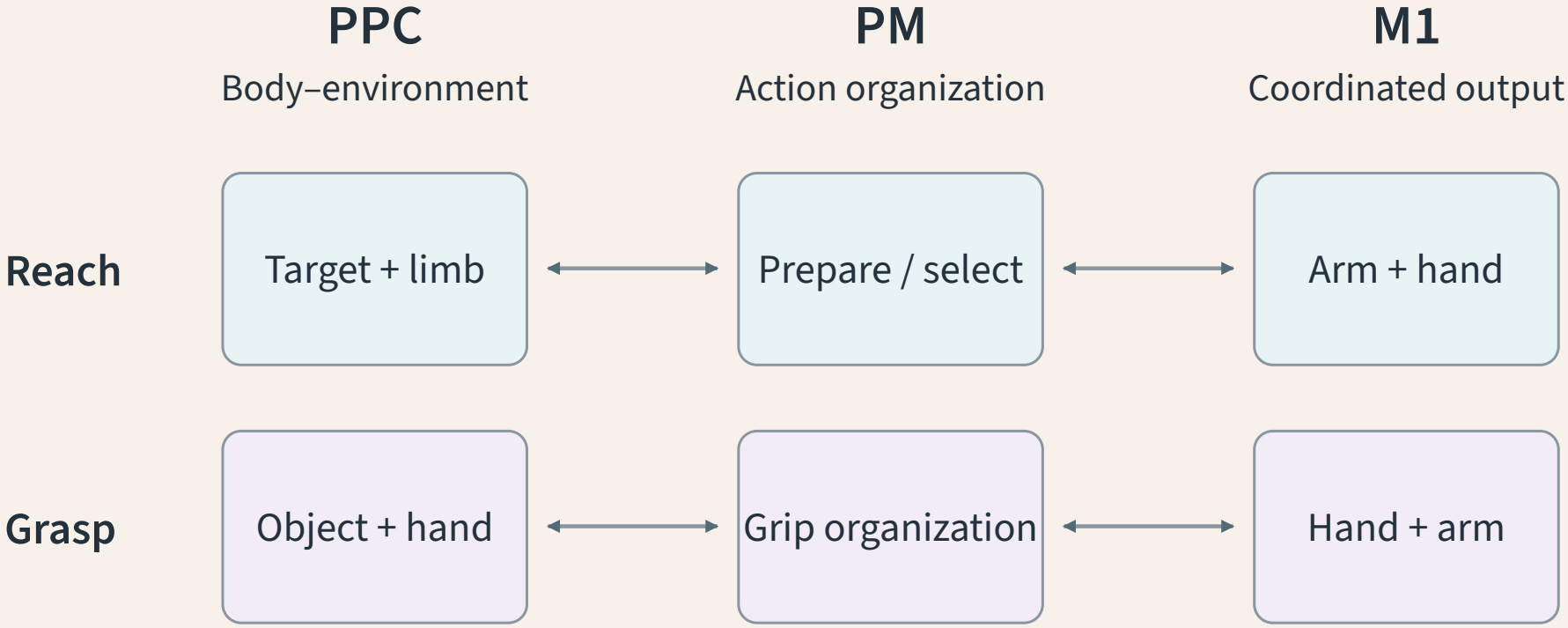
Kandel 6e, Figs. 34–16, p. 842; 34–18, p. 846; 34–22A/B, p. 850

E · Learning changes a connected controller



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

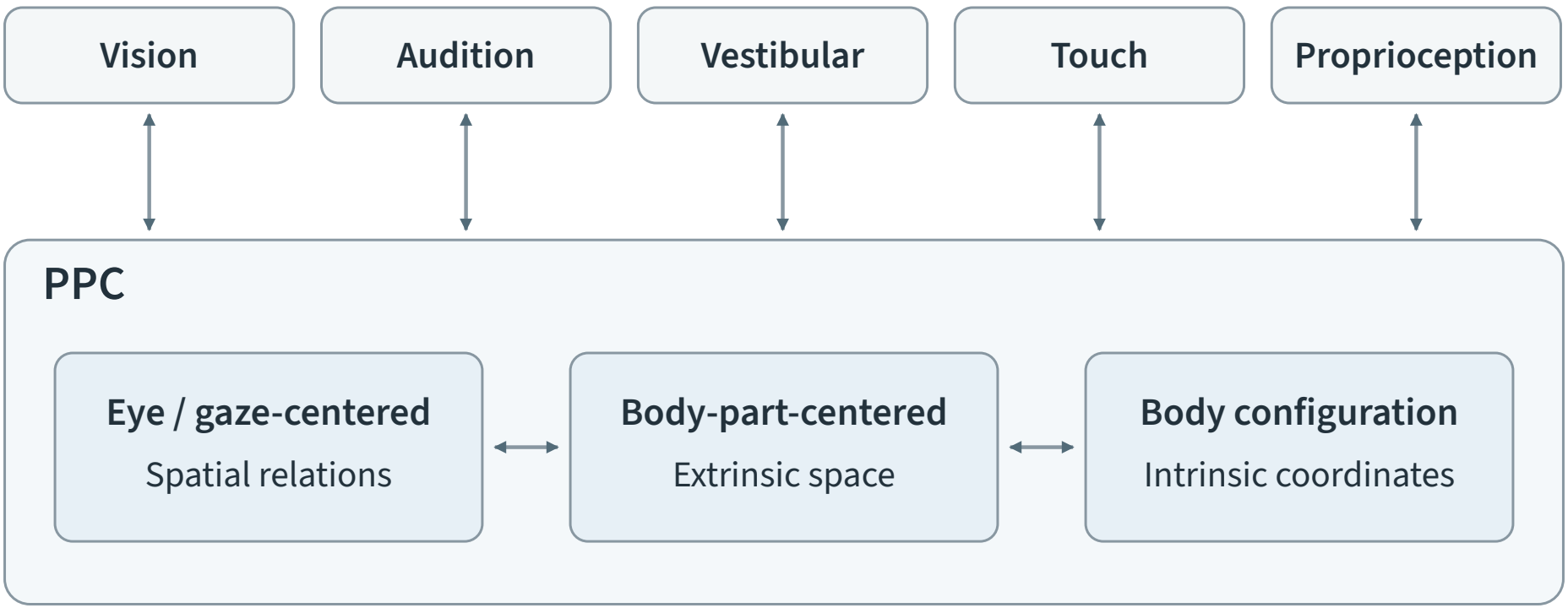
E · Learning changes a connected controller



Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

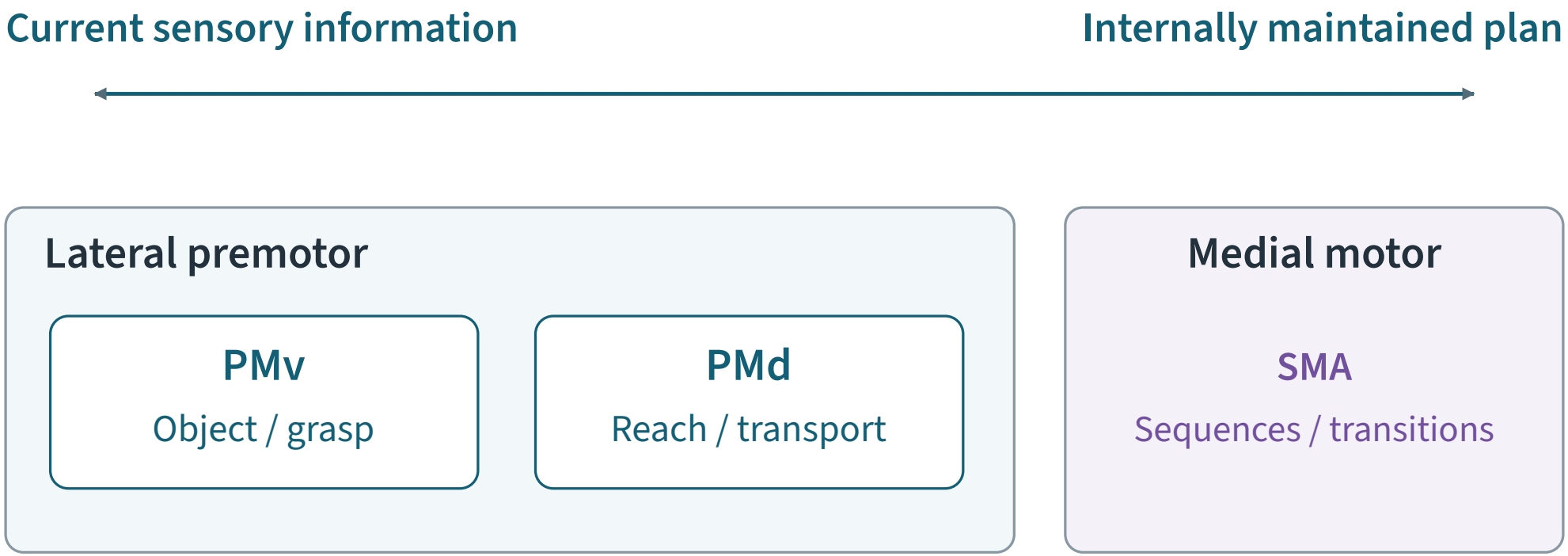
> Learning can change the connected controller developed in sections A–D.

Recall PPC's integrated body–environment relationships



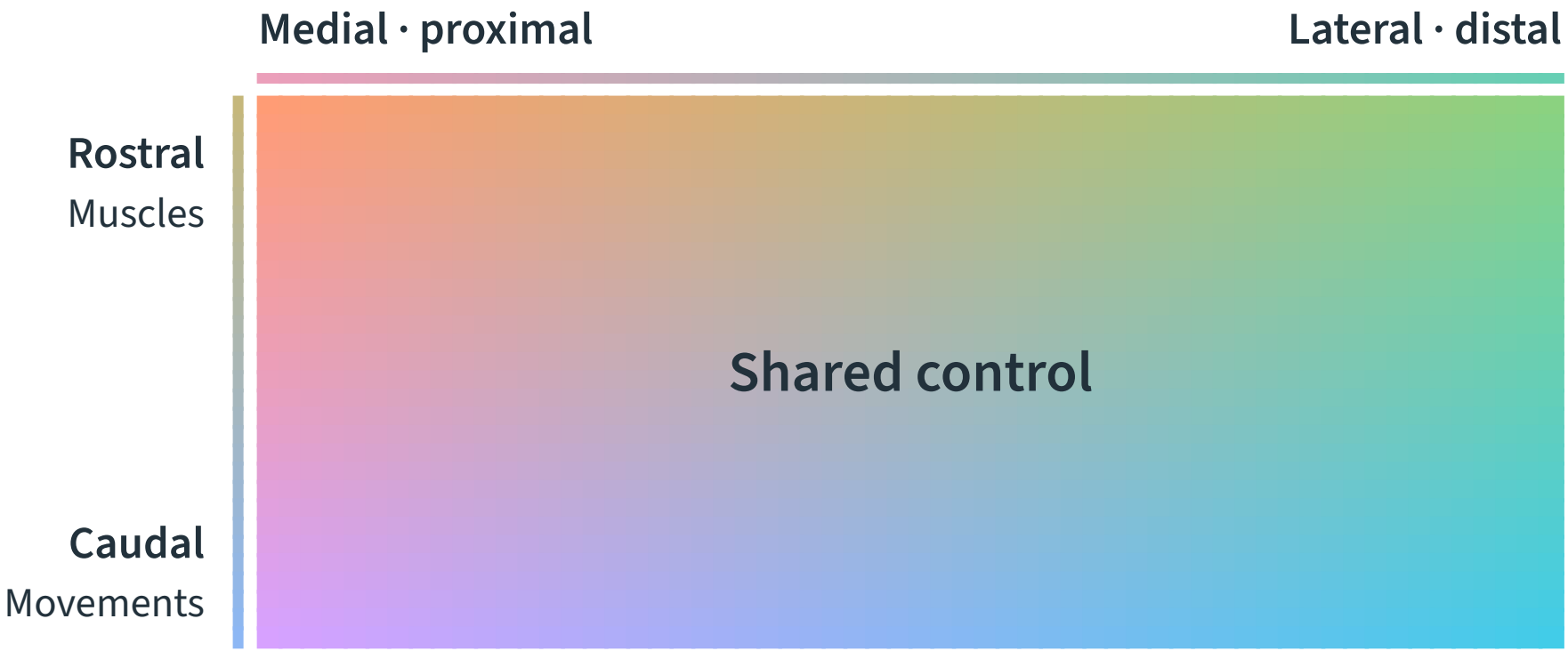
Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

Recall premotor action organization



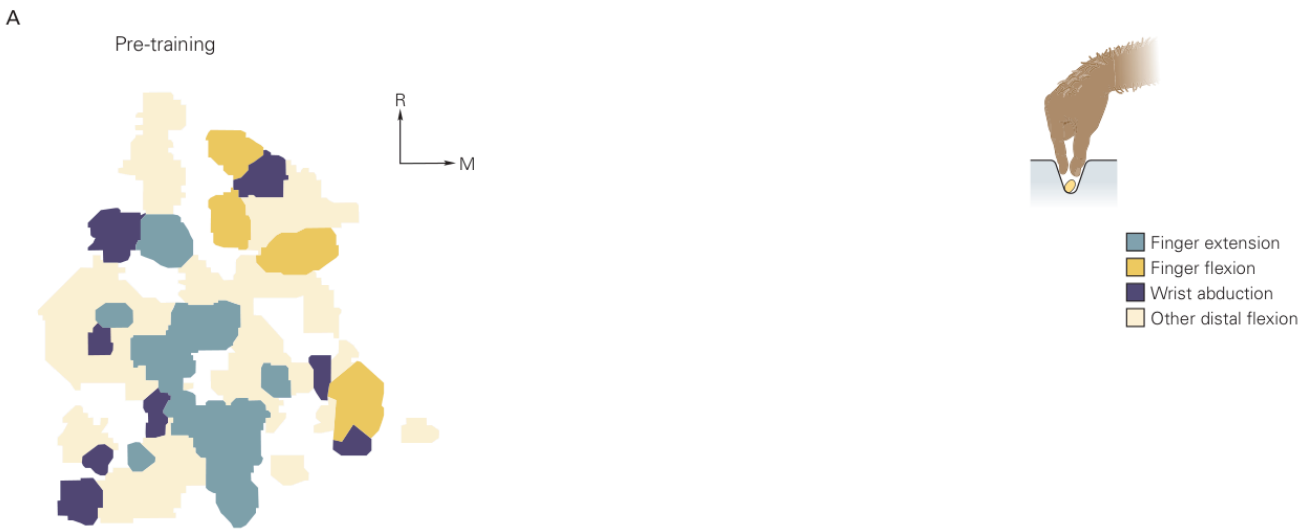
Kandel 6e, Ch. 34 · explanatory schematic, not measured wiring

Recall M1’s overlapping control

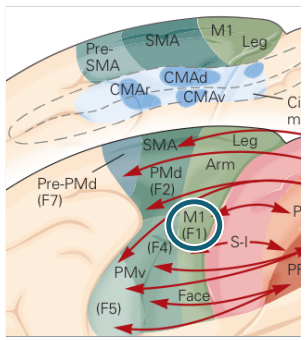


Kandel 6e, Ch. 34; Kolasinski et al. (2020) · conceptual sketch; retain the anatomy and coding distinctions

M1 map · practice changes the evoked movements



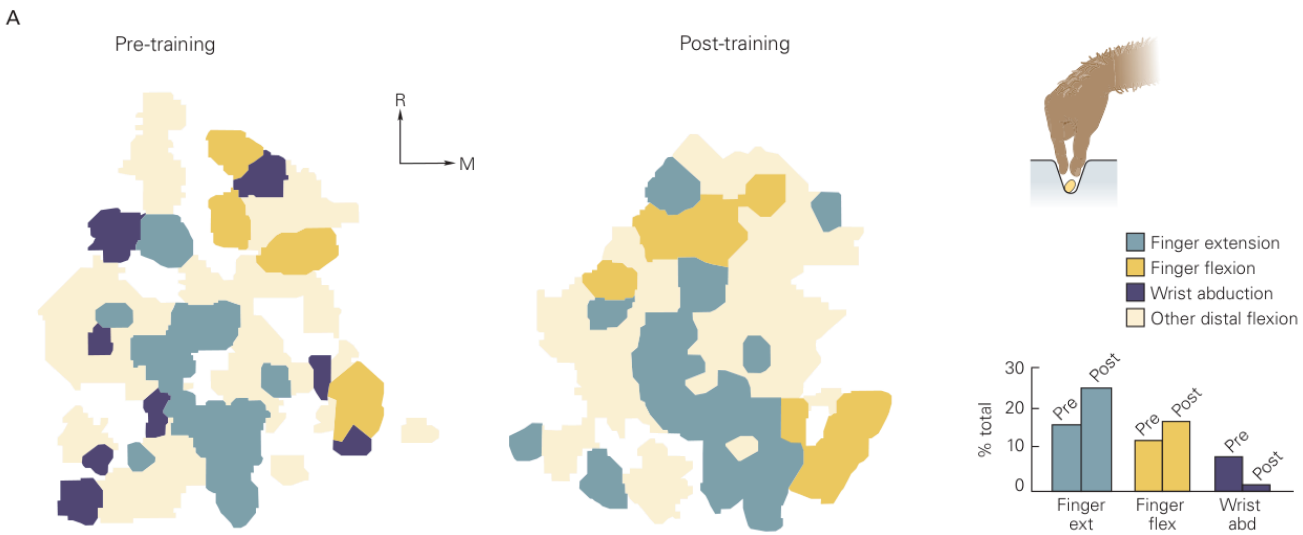
Primary motor cortex



- Training: retrieve pellets from a small well
- Map evoked movements, before training

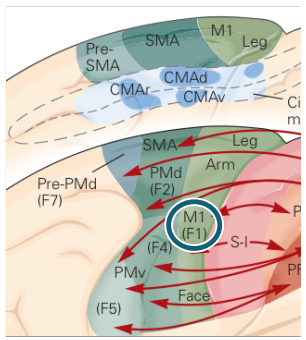
Kandel 6e, Fig. 34–24A, p. 854; 34–2B, p. 820 · pre-training and post-training maps

M1 map · practice changes the evoked movements



Kandel 6e, Fig. 34–24A, p. 854; 34–2B, p. 820 · pre-training and post-training maps

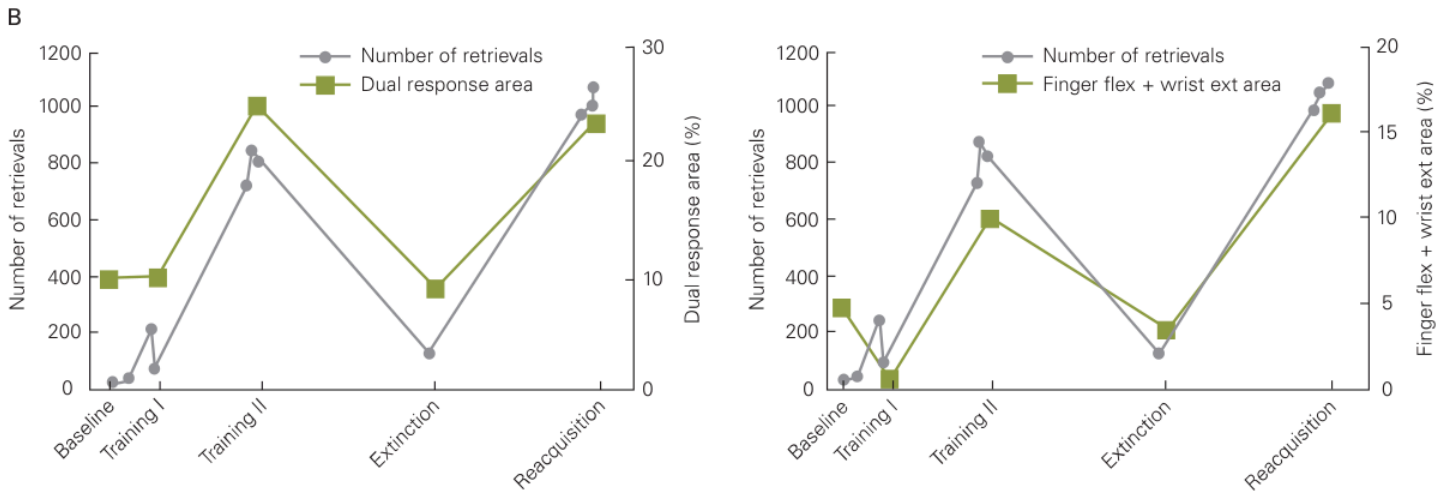
Primary motor cortex



- Training: retrieve pellets from a small well
- Map evoked movements, before training

- Finger areas expand
- Wrist abduction shrinks

M1 map · changes track skill acquisition and loss

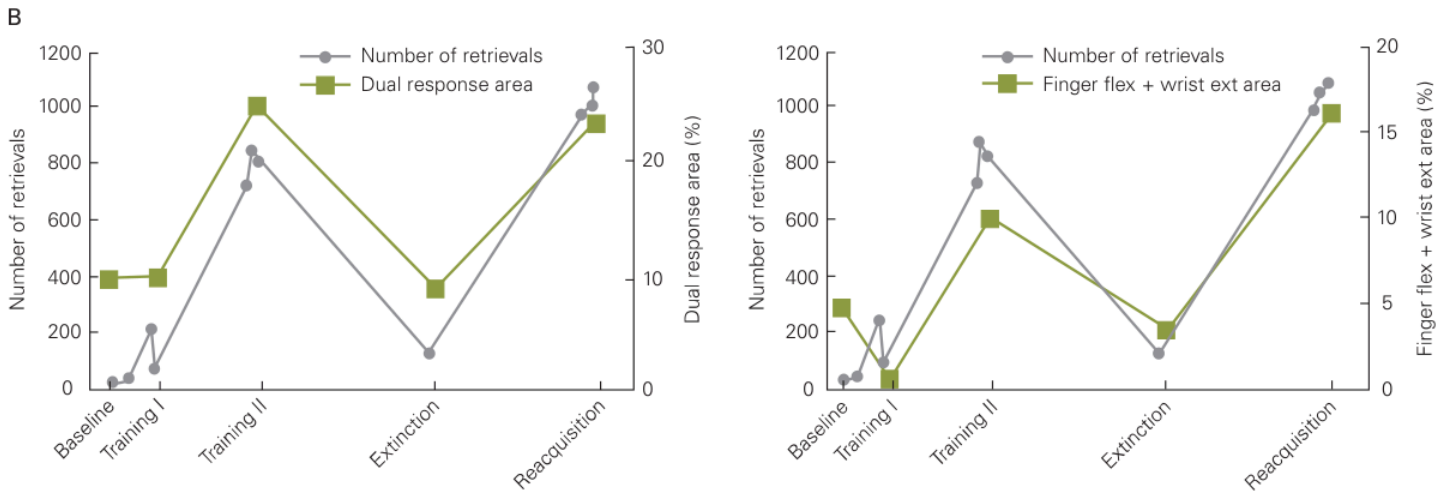


Primary motor cortex

- Retrievals and mapped area across training, extinction and retraining
- A different monkey from the maps

Kandel 6e, Fig. 34–24B, p. 854 · performance and mapped areas in a different monkey

M1 map · changes track skill acquisition and loss



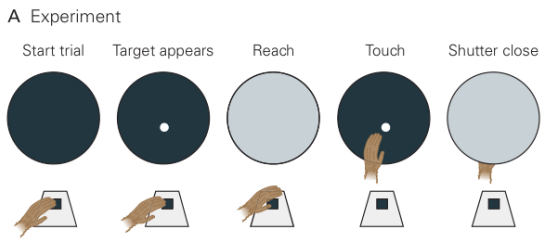
Primary motor cortex

- Retrievals and mapped area across training, extinction and retraining
- A different monkey from the maps

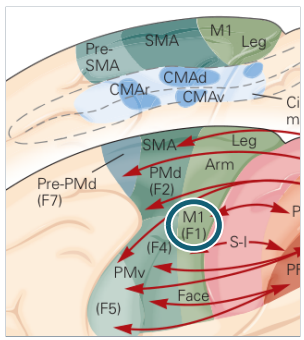
- Map area tracks performance
- Different measures, same course

Kandel 6e, Fig. 34–24B, p. 854 · performance and mapped areas in a different monkey

M1 · movement versus feedback about its result



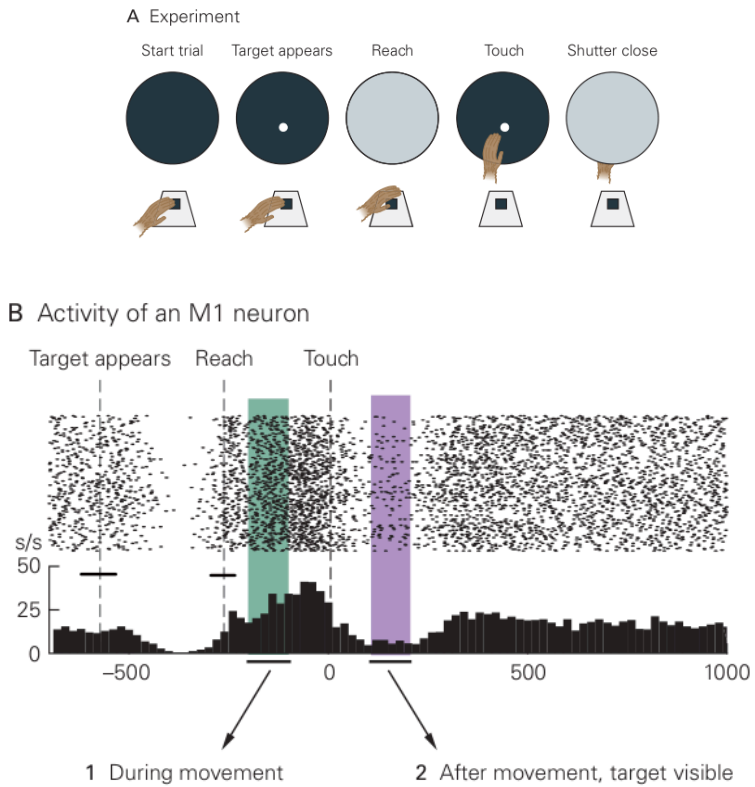
Primary motor cortex



- Prisms shift the seen target
- Shutter: no view during the reach
- Brief view of the endpoint after touch

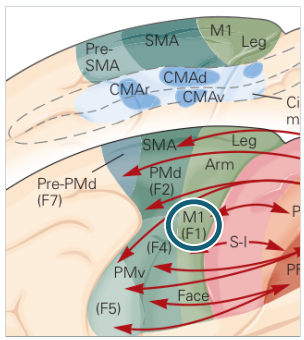
Kandel 6e, Fig. 34–25A/B, p. 856; 34–2B, p. 820 · prism shift, shutter and brief endpoint feedback; example M1 neuron

M1 · movement versus feedback about its result



Kandel 6e, Fig. 34–25A/B, p. 856; 34–2B, p. 820 · prism shift, shutter and brief endpoint feedback; example M1 neuron

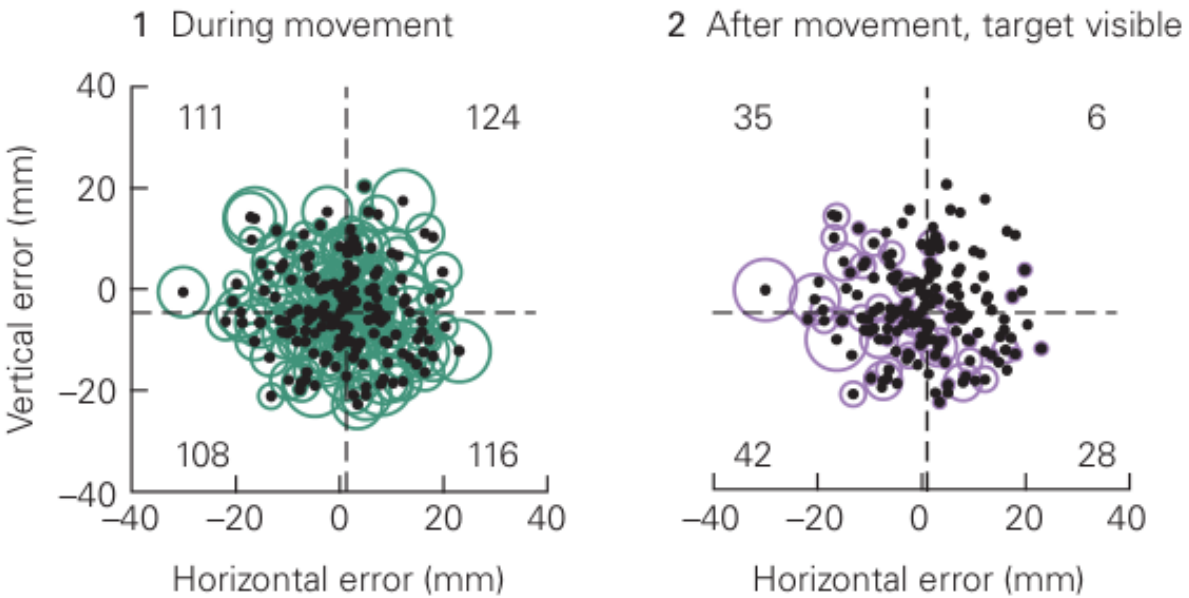
Primary motor cortex



- Prisms shift the seen target
- Shutter: no view during the reach
- Brief view of the endpoint after touch

> - Green: during movement
> - Purple: after feedback
> - Next: relate each to the error

M1 neuron · error direction after feedback

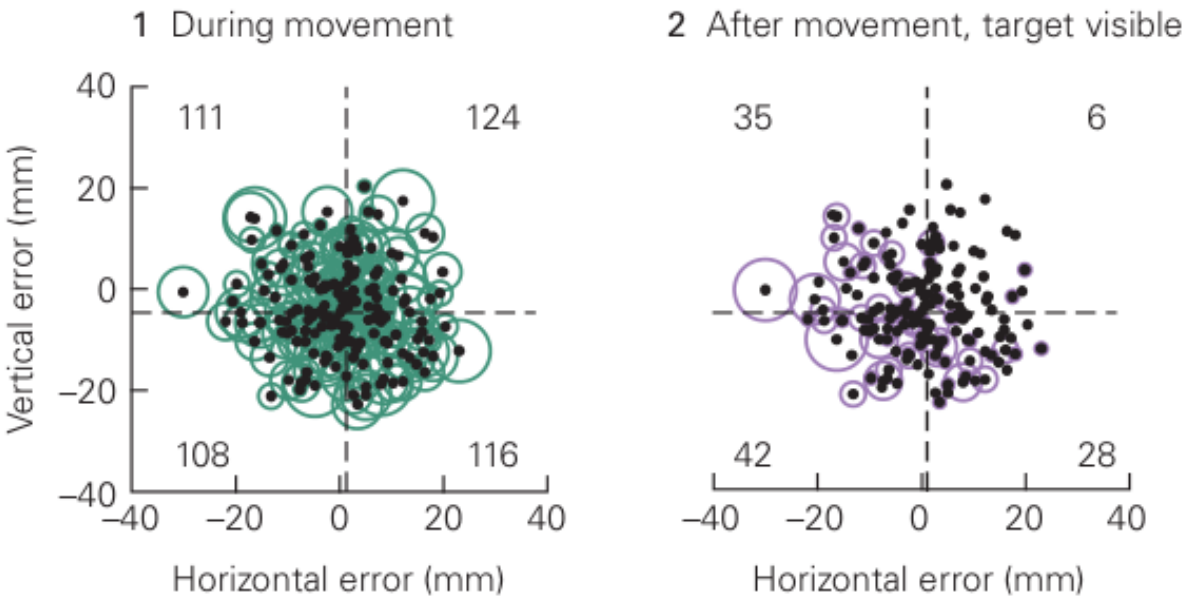


Kandel 6e, Fig. 34–25B1/B2, p. 856 · endpoint errors and firing-rate circles

Primary motor cortex

- Dots: endpoint errors from the target
- Circle size: firing in the epoch

M1 neuron · error direction after feedback



Primary motor cortex

- Dots: endpoint errors from the target
- Circle size: firing in the epoch

> - During movement: no relation
> - After feedback: error direction

Kandel 6e, Fig. 34–25B1/B2, p. 856 · endpoint errors and firing-rate circles

Testing a contribution to learning requires an intervention

OBSERVE

Error-related activity after feedback

INTERVENE

Simulate the activity with cortical stimulation

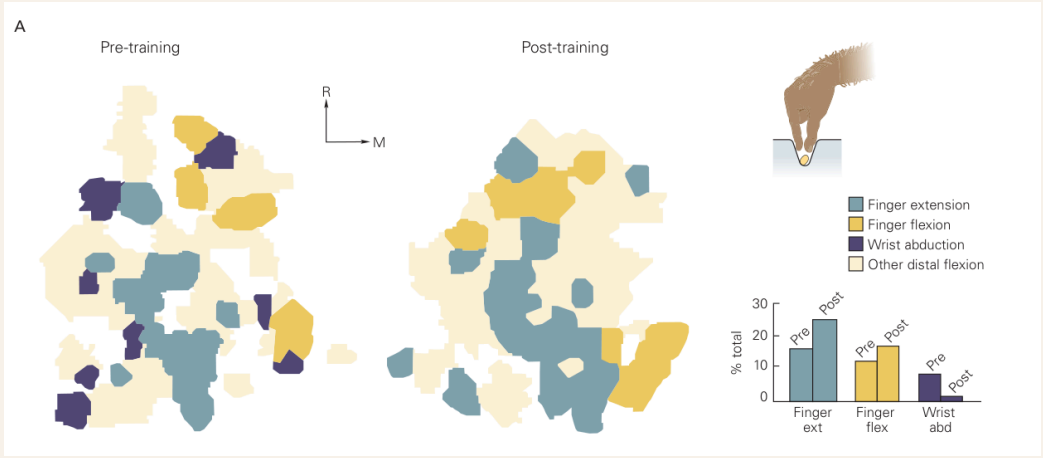
TEST

Does the next reach change?

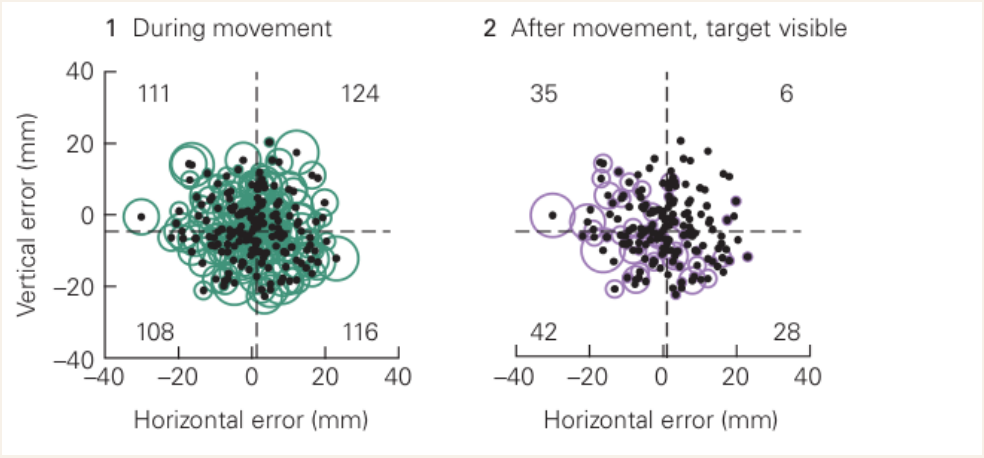
> The text reports adaptive changes even without an actual reaching error.

Kandel 6e, p. 855 · described M1/PMd intervention; separate from Fig. 34–25's recorded responses.

Section E summary: the controller changes with experience



Practice reshapes the map



Error direction after feedback

TAKE HOME Practice and error feedback change the cortical controller.

Kandel 6e, Figs. 34–24A, p. 854; 34–25B, p. 856